PROJECT DOCUMENTATION

CO_2 Monitor

ESP32-C6 · Sensirion SCD-41 CO_2 sensor · SHT45 temperature & humidity · 0.96" OLED

Version 0.20 – Step 3: web page, sleep mode, compact case v4 and its 3D model **Updated** 4 Nov 2026

Status breadboard prototype; case v4 designed

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1 Project overview

Step 1 goal: the OLED shows the temperature and humidity from the **SHT45** and the CO_2 concentration from the **SCD-41**. To save power and prevent OLED burn-in, the display switches off after **30 seconds**. A **push button** wakes it for another 30 seconds. The sensors keep measuring while the display is off, and the dimmed **RGB LED** on the ESP32 board shows the CO_2 level all the time as a colour from green through yellow to red.

Step 2: the monitor joins your Wi-Fi, gets the time from the internet and keeps a **history** of the values in its own flash memory: 1-minute values for 7 days, 5-minute values for 90 days and hourly values for about 3½ years (section 7).

Step 3 (this version): the monitor serves its own **web page**. Any browser on the home network shows the live values, graphs of any stored day and time range, and a CSV download, in light or dark mode (section 8). A **sleep mode** for the night switches the RGB LED off and shows a small moon on the display; a long press on the button or a switch on the web page turns it on and off (section 6). Next step: Matter for the Alexa Echo.

The ESP32 talks to all three modules over one shared **I²C bus**, made of four wires: 3.3 V, GND, SDA (data) and SCL (clock). Each module has its own fixed address, so they don't get in each other's way. The button is a separate input on GPIO2.

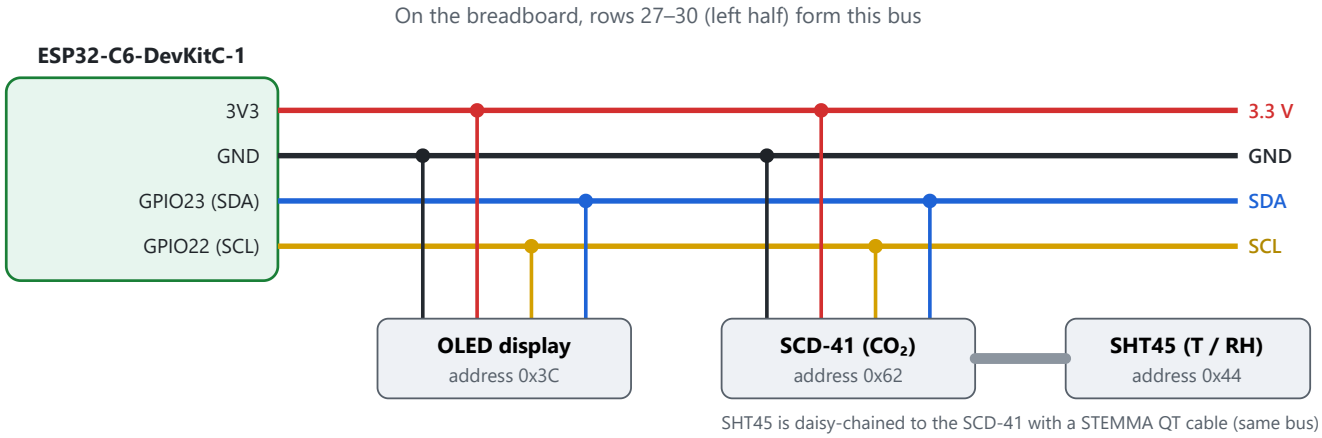


Figure 1 – Every module hangs on the same four wires. A dot means "connected"; lines that cross without a dot are not connected.

Project files

File	Purpose
`firmware/co2_monitor/co2_monitor.ino`	Main program (step 3): sensors → display, button wake, 30 s timeout, CO_2 colour on the RGB LED, sleep mode, serial output, Wi-Fi and internet time, Serial Monitor commands.
`firmware/co2_monitor/history.h`, `history.cpp`	History storage in the flash (section 7). The Arduino IDE shows them as tabs next to the main program.
`firmware/co2_monitor/web.h`, `web.cpp`	Web server: the page and the data addresses it uses (section 8).
`firmware/co2_monitor/web/index.html`, `make_web_page.ps1`, `../web_page.h`	The web page; the script packs it into `web_page.h`, the compressed copy the firmware serves (8.6).
`firmware/co2_monitor/secrets_example.h`	Template for `secrets.h`.
`firmware/co2_monitor/secrets.h`	Your Wi-Fi name and password (5.8). Not stored in git.
`firmware/i2c_scanner/i2c_scanner.ino`	Wiring check: lists all I^2C devices it finds. Run this first.
`firmware/scd41_calibration/scd41_calibration.ino`	Optional tool: forced recalibration (FRC) of the SCD-41 in fresh outdoor air.
`docs/CO2_Monitor_Documentation.html`	This document (source).
`docs/CO2_Monitor_Documentation.pdf`	PDF version, regenerated from the HTML after every change.
`docs/build_pdf.ps1`	Script that turns the HTML into the PDF (uses Chrome or Edge in headless mode).
`docs/web_page.png`	Screenshot of the web page (Figure 6).
`docs/case_v4_*.png`	Pictures of the case model (Figures 8–11).
`cad/adafruit/*.step`, `LICENSE`	Adafruit's 3D models of the sensor and speaker boards (MIT licence), the reference for the case model (section 13).
`cad/case_v4.py`, `viewer_template.html`, `requirements.txt`	The case model, design v4 (section 13): one Python script with every size. It writes the STEP files for the printer, STL files and a 3D viewer page into `cad/out/v4/`. The output folders and the Python environment `cad/.venv/` are not stored in git.
`cad/case.py`	The earlier case model, design v3 with the breadboard, kept for reference; writes into `cad/out/`.

2 Hardware list

#	Component	Role	Interface	Supply
1	**Espressif ESP32-C6-DevKitC-1-N8** ESP32-C6-WROOM-1-N8 module, 8 MB flash	Microcontroller. Reads the sensors, drives the display.	2× USB-C	5 V via USB → on-board 3.3 V regulator
2	**Adafruit SCD-41** (STEMMA QT) Sensirion SCD41	CO_2 sensor (photoacoustic NDIR)	I^2C `0x62`	3.3–5 V (we use 3.3 V)
3	**Adafruit SHT45** (STEMMA QT) Sensirion SHT45	Temperature and relative humidity	I^2C `0x44`	3–5 V (we use 3.3 V)

4	**Joy-it 0.96" OLED** 128×64 (SBC-OLED01) SSD1306 controller	Display	I²C `0x3C` (or `0x3D`)	3.3–5 V (we use 3.3 V)
5	STEMMA QT → male header pins cable (4-pin)	Connects the breadboard bus to the SCD-41	JST SH 1.0 mm	–
6	STEMMA QT ↔ STEMMA QT cable	Daisy-chains SHT45 to the SCD-41	JST SH 1.0 mm	–
7	4-pin tactile push button	Wakes the display	GPIO2	–
8	400-point breadboard	Carries the ESP32, OLED, button and the bus	–	–
9	Male–male jumper wires, 5 (one of them long) (needed, not in the original list)	Breadboard connections	–	–
10	USB-C **data** cable (needed)	Power and programming. Charge-only cables don't work.	–	–

Planned: more hardware will be added in later steps. Add new I²C devices to the same bus (check that their address is not already used); give anything else its own free GPIO from the pin diagram.

3 ESP32-C6-DevKitC-1 pin diagram

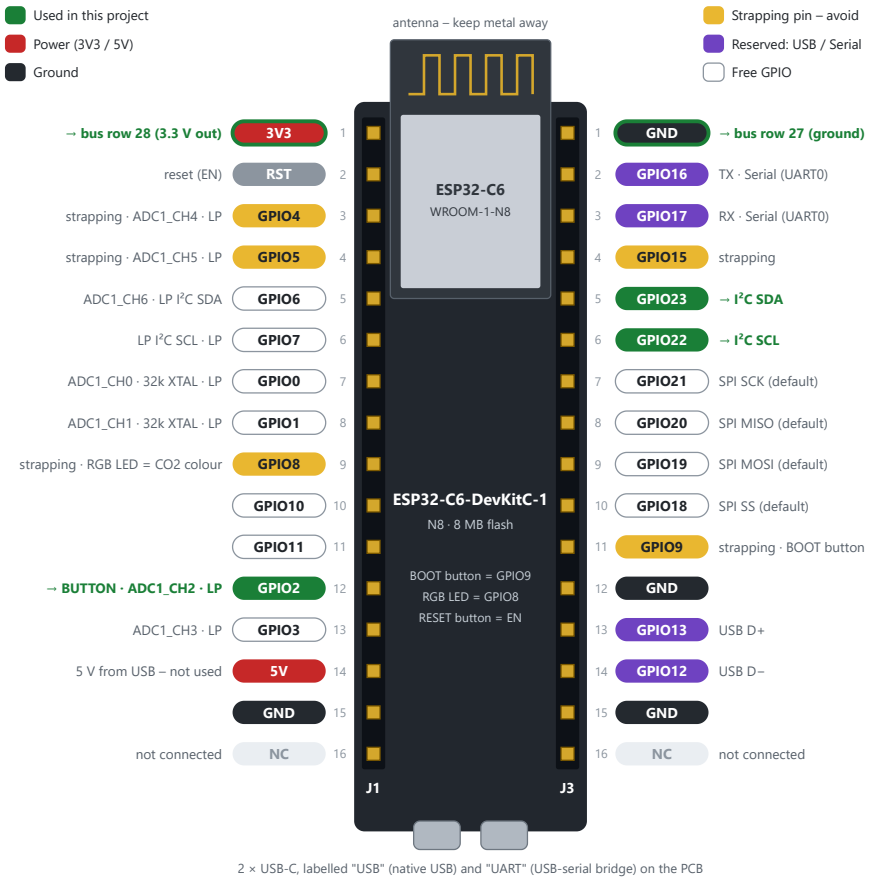


Figure 2 – Top view, antenna up, USB down. Green = used in this project. "LP" = low-power GPIO (can wake the chip from deep sleep). The drawing is schematic, not to scale; the labels printed on your board are what counts.

Pins used in step 1

Header pin	Signal	Used for	Breadboard hole → goes to
J1-1	**3V3**	3.3 V supply for OLED, SCD-41, SHT45	j18 → c28 (3.3 V bus row)
J1-12	**GPIO2**	Push button (internal pull-up, pressed = LOW)	j7 → j25 (button leg f25)
J3-5	**GPIO23**	I²C SDA (Arduino default for ESP32-C6)	a14 → c30 (SDA bus row)
J3-6	**GPIO22**	I²C SCL (Arduino default for ESP32-C6)	a13 → c29 (SCL bus row)
J3-1	**GND**	Ground	a18 → c27 (GND bus row)
(J1-9)	**GPIO8**	On-board RGB LED (CO_2 colour)	no wire, keep i10 empty

Pins to avoid (or use with care)

- **Strapping pins GPIO4, GPIO5, GPIO8, GPIO9, GPIO15:** their level during reset decides how the chip boots. A button or sensor pulling one of them low can stop the board from starting. GPIO8 also drives the on-board RGB LED, and GPIO9 is the BOOT button.
- **GPIO12 / GPIO13:** native USB (D− / D+). Using them breaks the "USB" port.
- **GPIO16 / GPIO17:** UART0 TX/RX, used by the "UART" USB port and the Serial Monitor.
- **5V pin:** carries 5 V from USB. **The GPIOs are not 5 V tolerant.** Never connect 5 V to a GPIO or to the breadboard bus.
- **Good free pins for later:** GPIO0, 1, 3, 6, 7 (low-power GPIOs, some with ADC), GPIO10, 11, 18, 19, 20, 21.

4 Wiring

Breadboard holes are named **column letter + row number**, e.g. `j19` = column j, row 19. On a 400-point breadboard, each row's five holes a–e are connected together, and so are f–j (but not across the centre gap). The long power rails run along the edges.

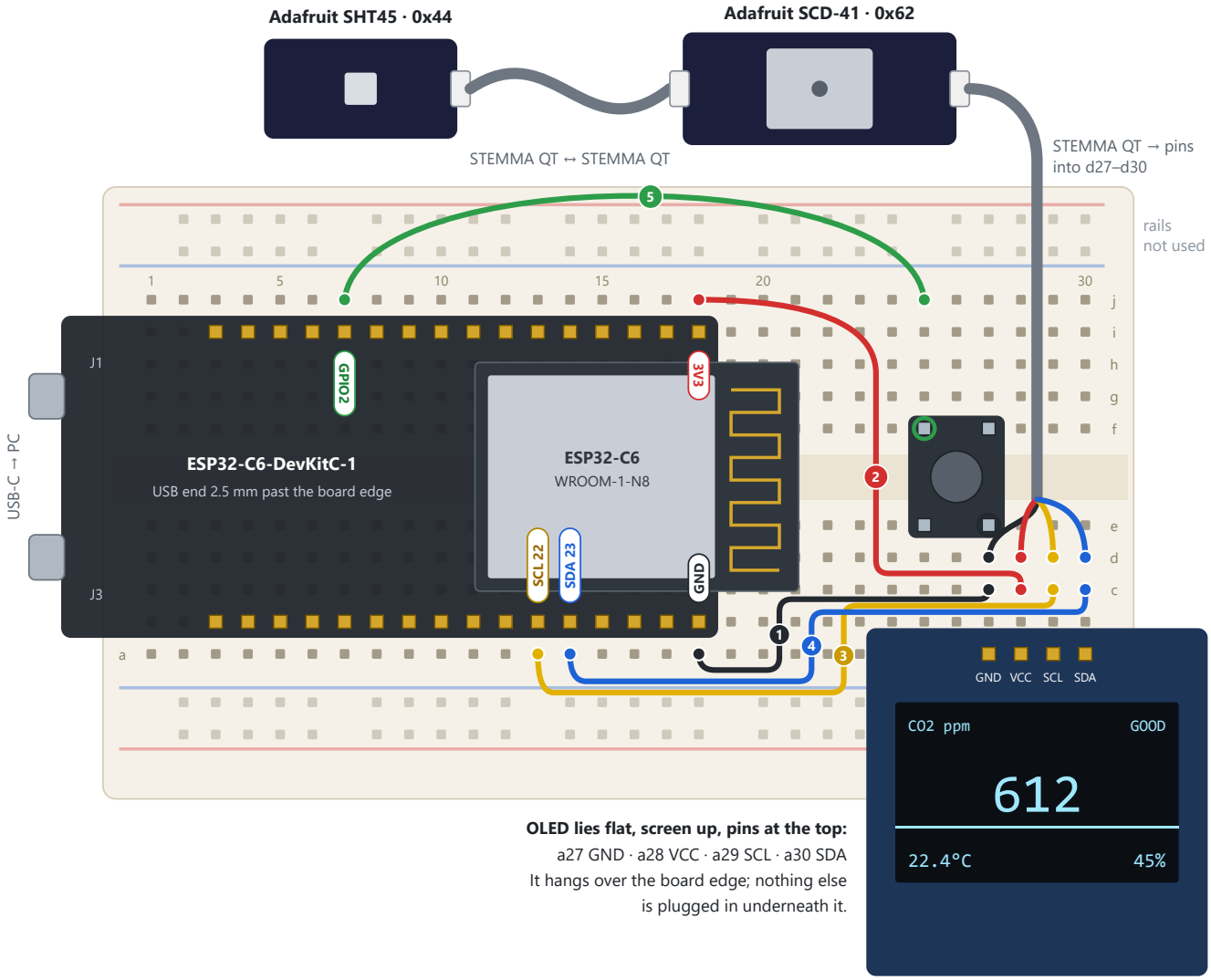


Figure 3 – Breadboard layout (top view, seen from the column-a side). The ESP32 sits in rows 3–18 (3V3 pin in i18) with its USB end about 2.5 mm past the board edge, so the USB-C ports reach just past the joining nubs; the antenna overhangs rows 19–21. Rows 27–30, left half, form the bus in the same order as the OLED's pins: **row 27 = GND, row 28 = 3.3 V, row 29 = SCL, row 30 = SDA**. The OLED lies flat in a27–a30 and hangs over the board edge, the ESP32 wires go into column c, the SCD-41 cable into column d. The button's GND leg (e27) sits directly in the GND row. The power rails are not used.

Connection list

#	Wire	From	To	Signal
1	black	a18 (J3-1 GND)	c27	GND → bus row 27
2	red (long)	j18 (J1-1 3V3)	c28	3.3 V → bus row 28
3	yellow	a13 (J3-6 GPIO22)	c29	SCL → bus row 29
4	blue	a14 (J3-5 GPIO23)	c30	SDA → bus row 30
5	green	j7 (J1-12 GPIO2)	j25	Button, leg f25
–	OLED header	a27 GND · a28 VCC · a29 SCL · a30 SDA (lying flat, pins at the top)		plugs straight into the bus rows

–	STEMMA QT → pins	d27 black (GND) · d28 red (V+) · d29 yellow (SCL) · d30 blue (SDA); other end into the SCD-41	SCD-41 on the bus
–	STEMMA QT ↔ STEMMA QT	SCD-41 second connector → SHT45	SHT45 on the bus
–	Button	legs in e25, e27, f25, f27; leg e27 sits in the GND row, so it needs no wire of its own	GPIO2 ↔ GND

Each bus row holds four things: the OLED pin (column a), the ESP32 wire (column c), the SCD-41 cable pin (column d) and, in row 27, the button leg (column e). The power rails are not used in this layout.

Step by step (USB unplugged!)

1. **ESP32:** pins in rows 3–18, so the USB-C ports stick out about 2.5 mm past the end of the breadboard (about 1 mm past its joining nubs, which the case needs), antenna overhanging rows 19–21, the pin labelled **3V3** in **i18**. One free hole is left next to every pin: column **j** on the J1 side, column **a** on the J3 side.
2. **OLED:** plug its header into **a27–a30** so that you can read the screen with the pins at the top: **GND → a27, VCC → a28, SCL → a29, SDA → a30**. The module lies flat, screen up, and hangs over the edge of the breadboard. Nothing else is plugged in underneath it.
3. **ESP32 to the bus:** wire 1 (a18 → c27, GND), wire 3 (a13 → c29, SCL), wire 4 (a14 → c30, SDA). Then wire 2 (j18 → c28, 3.3 V); it is the long one that crosses the board just past the antenna end.
4. **SCD-41:** plug the STEMMA QT → pins cable into one of its connectors. Put the pins in d27 (black), d28 (red), d29 (yellow), d30 (blue): the same order as the OLED.
5. **SHT45:** connect it to the SCD-41's second connector with the STEMMA QT ↔ STEMMA QT cable. Place it as far from the ESP32 and the SCD-41 as the cable allows.
6. **Button:** place it across the centre gap in rows 25–27, legs in e25, e27, f25, f27. Then wire 5 (j7 → j25). The button's leg e27 is in the GND row and its diagonally opposite leg f25 is wired to GPIO2. Diagonal legs work whichever way round the button's internal leg pairs are.
7. **Check before plugging in USB:** row 27 (GND) and row 28 (3.3 V) are not connected to each other; nothing is connected to the 5V pin (J1-14, hole i5); the STEMMA colours are in the right rows.

No pins pointing out of the back of your OLED, or not enough room? Connect it with four female–male jumper wires to the same rows instead (GND → row 27, VCC → 28, SCL → 29, SDA → 30) and put it wherever you like.

> **Power the sensors from 3.3 V only.** The Adafruit boards would accept 5 V, but their I²C pull-up resistors connect to their supply pin. At 5 V they would pull SDA/SCL to 5 V, and the ESP32-C6 pins are not 5 V tolerant.

5 Arduino IDE setup

> **Already done on this PC:** Arduino IDE 2 is installed, and so is the board package **esp32 by Espressif Systems 3.3.12**. You still need to install the libraries (5.3) and the CP210x USB driver for the "UART" port (5.5).

5.1 Arduino IDE

Download **Arduino IDE 2.x** from `arduino.cc/en/software` and install it with the default options. Allow the USB driver installation if Windows asks.

5.2 ESP32 board package

1. **Tools → Board → Boards Manager…**, search `esp32`.

2. Install **"esp32 by Espressif Systems"**, version 3.x. (Not "Arduino ESP32 Boards", which is a different package for Arduino's own boards.)
3. If it isn't listed: **File → Preferences → Additional boards manager URLs**, add `https://espressif.github.io/arduino-esp32/package_esp32_index.json`, then search again.

5.3 Libraries

Sketch → Include Library → Manage Libraries... (or the books icon on the left). For each library, click **INSTALL**. When the IDE asks about dependencies, choose **INSTALL ALL**.

Search for	Install	Pulls in automatically	Tested version
`Adafruit SSD1306`	Adafruit SSD1306 (by Adafruit)	Adafruit GFX Library, Adafruit BusIO	2.5.17 / GFX 1.13.0 / BusIO 1.17.4
`Adafruit SHT4x`	Adafruit SHT4x Library (by Adafruit)	Adafruit Unified Sensor (+ Adafruit SH110X, unused)	1.0.5 / Unified Sensor 1.1.15
`Sensirion I2C SCD4x`	Sensirion I2C SCD4x (by Sensirion)	Sensirion Core	1.1.0 / Core 0.7.3

> Use **Sensirion I2C SCD4x version 1.x**. Version 0.4 and older have a different API (class `SensirionI2CScd4x`), and the sketches won't compile with it.

5.4 Board settings (Tools menu)

Setting	Value
Board	**ESP32C6 Dev Module** (Tools → Board → esp32)
Port	the COM port that appears when you plug in the board
USB CDC On Boot	**Disabled** if you use the port labelled **UART** · **Enabled** if you use the port labelled **USB**
Flash Size	**8MB (64Mb)**
Partition Scheme	**8M with spiffs (3MB APP/1.5MB SPIFFS)**. The history lives in its 1.5 MB data area (section 7).
CPU Frequency	160MHz (WiFi)
Upload Speed	921600 (use 115200 if uploads fail)
Erase All Flash Before Sketch Upload	**Disabled** (the default). Enabled deletes the stored history.
Everything else	leave at default

> **Keep Flash Size and Partition Scheme as shown.** Normal uploads keep the stored history. Changing either setting moves the data area, so the history can't be read any more. The firmware checks the layout at every start: with a different scheme it prints `History OFF: wrong partition scheme` and records nothing.

5.5 Which USB port?

Both USB-C ports can power and program the board. **Recommended: the port labelled "UART"** with *USB CDC On Boot = Disabled*. Serial output then appears from the first moment after reset. If you use the "USB" port, set *USB CDC On Boot = Enabled*, otherwise the Serial Monitor stays empty.

Drivers: the "USB" port uses a driver built into Windows and works straight away. The "UART" port goes through a **Silicon Labs CP2102N** USB-to-UART chip, and **on this PC Windows had no driver for it**. Device Manager listed

"CP2102N USB to UART Bridge Controller" under *Other devices* with error *Code 28*, and no COM port appeared. Windows Update doesn't install this driver automatically for this chip, so install it once by hand:

1. Download **CP210x Universal Windows Driver** (`CP210x_Universal_Windows_Driver.zip`) from `silabs.com/developer-tools/usb-to-uart-bridge-vcp-drivers` , tab *Downloads*.
2. Extract the ZIP completely. Installing from inside the ZIP fails.
3. Device Manager → right-click "CP2102N USB to UART Bridge Controller" → **Update driver** → **Browse my computer for drivers** → select the extracted folder → **Next**. Alternatively, right-click `silabser.inf` → *Show more options* → **Install**.
4. The device moves to *Ports (COM & LPT)* as **"Silicon Labs CP210x USB to UART Bridge (COM*n*)"**. Select that COM*n* in the Arduino IDE. If it still shows the warning, unplug the board, plug it back in, or restart the PC.

If no device appears at all, not even under *Other devices*, try another cable: many USB-C cables are charge-only.

5.6 First upload test

1. Plug in the board, select board and port as above.
2. **File** → **Examples** → **ESP32** → **GPIO** → **BlinkRGB** and click **Upload** (→ arrow).
3. The on-board RGB LED should cycle through colours. The toolchain works.
4. If the upload says "*Failed to connect*": hold `BOOT` , tap `RESET` , release `BOOT` , then upload again. Press `RESET` after the upload finishes.

5.7 Open the project sketches

File → **Open...** and pick the `.ino` file inside `firmware/<name>/` . The Arduino IDE needs each sketch in a folder with the same name, which is how the project is laid out. Serial Monitor: **Tools** → **Serial Monitor**, set to **115200 baud**.

5.8 Wi-Fi name and password (`secrets.h`)

The monitor needs Wi-Fi for the clock, and later for the web page. Your network name and password go in `firmware/co2_monitor/secrets.h` . Git ignores this file, so the password never ends up in the repository. The IDE shows it as a tab next to `co2_monitor.ino` .

1. If `secrets.h` is missing (for example in a fresh copy of the project), copy `secrets_example.h` and rename the copy to `secrets.h` . Without it the sketch doesn't compile.
2. Replace `your-wifi-name` and `your-wifi-password` . Keep the quotes.
3. The ESP32-C6 only uses **2.4 GHz** Wi-Fi. A router that sends 2.4 and 5 GHz under one name is fine.

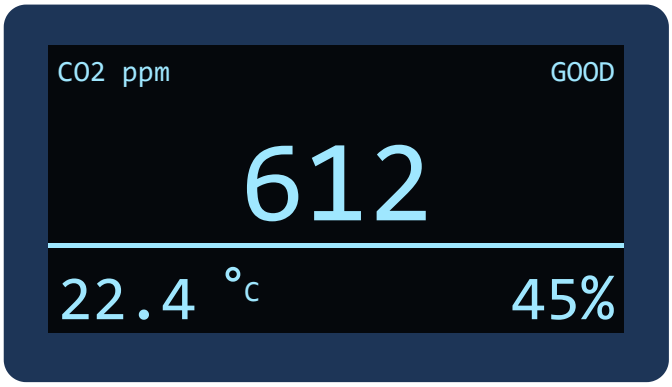
```
constexpr char WIFI_SSID[] = "MyNetwork";
constexpr char WIFI_PASSWORD[] = "my secret password";
```

The monitor shows up in your router's device list as **co2-monitor**. For the web page, reserve a fixed address for it in the router later (DHCP reservation).

6 Firmware (`co2_monitor.ino`)

What it does

- Starts the I²C bus on GPIO23 (SDA) / GPIO22 (SCL).
- **SHT45**: high-precision reading every 2 s, heater off.
- **SCD-41**: periodic measurement, a new CO_2 value every 5 s. Altitude compensation is set at every start.
- **Warm-up**: for the first 30 s after start-up the sensors measure, but no value is shown, printed or saved. The values from this time are thrown away. The display counts down ("warming up", 30 s … 1 s).
- **Display**: on during the warm-up, then for 30 s with the first values, and for 30 s after every button press. Pressing while it's on restarts the 30 s. When off, the panel is fully dark and the content is kept. Brightness is set by `OLED_BRIGHTNESS`.
- **RGB LED** (on the board, GPIO8): dimmed, on all the time except in sleep mode, shows the CO_2 level as a colour that changes smoothly from green through yellow to red. It stays off until the first CO_2 value after the warm-up.
- **Sleep mode**: RGB LED off, a small moon on the display. Hold the button for 2 s or use the switch on the web page (see *Sleep mode* below).
- **Serial**: one line per CO_2 reading after the warm-up, usable in the Serial Monitor and the Serial Plotter.
- **Wi-Fi and clock**: connects with the details in `secrets.h` and reconnects by itself. Gets the time (UTC) from the internet and updates it every 3 hours.
- **History**: stores 1-minute, 5-minute and hourly values in the flash (section 7). You can view them with commands in the Serial Monitor (7.6).
- **Web page**: live values, graphs and CSV downloads in a browser on the home network (section 8).
- A missing module is reported on Serial and on the display (`--.-` / "SCD-41 missing"). The rest keeps working.



Sleep mode on: moon at the top After a long press, for 2.5 s

Figure 4 – Display layout: CO_2 and rating on top, temperature and humidity (from the SHT45) at the bottom. In sleep mode a small moon sits between "CO2 ppm" and the rating. A long press on the button confirms the new mode on the full screen (a sun with "off - LED on" when it is switched off).

Settings at the top of the sketch

Constant	Default	Meaning
`SENSOR_ALTITUDE_M`	0	**Set this to your altitude in metres above sea level.** The SCD-41 uses it for air-pressure compensation. Without it, readings at e.g. 500 m are a few percent off.
`WARMUP_TIME_MS`	30000	Warm-up after start-up (ms): the sensors measure, but nothing is shown, printed or saved.

`OLED_ADDRESS`	0x3C	0x3D if the address resistor on the back of the OLED was moved.
`OLED_ROTATION`	0	0 = normal, header pins at the top. Set 2 if the text appears upside down (e.g. if you mount the display the other way round).
`OLED_BRIGHTNESS`	10	Display brightness 0–255 (the SSD1306's contrast setting; 207 is the library default). The range is smaller than it sounds: even 0 stays readable, at roughly half brightness. Lower values also slow burn-in.
`DISPLAY_ON_TIME_MS`	30000	How long the display stays on (ms).
`LONG_PRESS_MS`	2000	How long the button must be held to switch sleep mode on or off (ms).
`SLEEP_MESSAGE_MS`	2500	How long the display confirms the new mode after a long press (ms).
`BUTTON_PIN`	2	Button GPIO (other leg to GND).
`I2C_SDA_PIN` / `I2C_SCL_PIN`	23 / 22	I^2C pins.
`CO2_GOOD_BELOW`, `CO2_OK_BELOW`, `CO2_VENTILATE_BELOW`	800 / 1000 / 2000	Limits for the rating shown at the top right, and the points where the LED is green, yellow and red.
`LED_BRIGHTNESS`	5	Brightness of the RGB LED, 1–255 (0 = off). 255 is glaring; the BlinkRGB example uses 64. Try 3–5 for a bedroom, 20–30 for a bright room.
`LED_GREEN`, `LED_YELLOW`, `LED_RED`	{0,255,0} / {255,160,0} / {255,0,0}	The three colours at full brightness, as {red, green, blue}. Yellow has less green because the green LED looks brighter than the red one. If yellow looks too green, lower its middle value.
`WIFI_HOSTNAME`	co2-monitor	The name shown in your router's device list.
`NTP_SERVER_1`, `NTP_SERVER_2`	pool.ntp.org / time.google.com	Internet time servers. Your router's address also works if it offers time itself.

The Wi-Fi name and password are in `secrets.h` (5.8). The history settings are in `history.cpp` (7.9).

CO_2 rating on the display and the LED

CO_2 (ppm)	Display	LED	Meaning
below 800	**GOOD**	green	Well-ventilated room (outdoor air is about 420–430 ppm).
800 – 999	**OK**	green → yellow	Acceptable.
1000 – 1999	**VENTILATE** (inverted)	yellow → orange → red	Air is getting stale. Open a window. 1000 ppm is the classic guideline (Pettenkofer value).
2000 and above	**BAD AIR** (inverted)	red	Ventilate now. Tiredness and poor concentration are common at these levels.

The LED colour changes smoothly with the CO_2 value; for example, 1500 ppm is orange. It is updated with every new reading (every 5 s) and stays on while the display is off. In sleep mode it is off.

Sleep mode

For the night. Sleep mode switches the **RGB LED off** and puts a **small moon** in the top line of the display (Figure 4). Everything else carries on: the sensors measure, the history records, and the web page works. Sleep mode is **not** the ESP32's deep sleep, which would stop all of that.

Switch it with	How
The button	**Hold it for 2 s.** The display wakes as soon as you press, and after 2 s it shows a big moon with "Sleep mode on - LED off" (or a sun with "off - LED on") for 2.5 s. When switching on, the display then goes dark at once; when switching off, it shows the values for the rest of the 30 s. A short press still just wakes the display, in sleep mode with the moon at the top.
The web page	The *Sleep mode* switch at the top right (8.3). The display stays dark, so the bedroom does too; if it happens to be on, the moon appears or disappears.
The Serial Monitor	`sleep on` or `sleep off` (7.6). `status` shows the current mode.

- **Remembered:** the mode is stored in the ESP32's settings memory (NVS), apart from the history. It survives a restart, a power cut and a firmware upload. *Erase All Flash* resets it to off.
- **Lights that stay on:** the ESP32 board's red power LED and the green "on" LEDs of the SCD-41 and SHT45 are wired straight to the power supply. No software can switch them off. How to cover or disconnect them: section 11.
- After a restart in sleep mode the display still shows the warm-up and the first values, as always (about 1 minute), then stays dark.

7 History storage

The monitor keeps a long-term record of its values in the ESP32's own flash memory, with no extra hardware. The web page (section 8) draws its graphs from it and offers it as a CSV download. You can also read it in the Serial Monitor (7.6). The code is in `history.h` and `history.cpp`.

7.1 Three levels of detail

Level	Each record holds	Record	Files	Kept
1-minute (`m1`)	average CO_2, temperature, humidity and air pressure	12 bytes	`/m1/20261028.bin`, one per day, up to 17 KB	7 days
5-minute (`m5`)	the same	12 bytes	`/m5/20261028.bin`, one per day, up to 3.5 KB	90 days
Hourly (`h1`)	average, minimum and maximum of CO_2, temperature and humidity; average pressure; number of minutes measured	26 bytes	`/h1/202610.bin`, one per month, up to 19 KB	until the flash is full, about 3½ years; then the oldest month is deleted

- Air pressure stays empty until the BMP390 is fitted. The light level isn't stored; the VEML7700 is meant for the display brightness.
- The hourly minimum and maximum come from the 1-minute averages, not from single 5-second readings.
- Each file starts with a format version, so a later firmware can add values without losing the older data.

7.2 How a reading gets saved

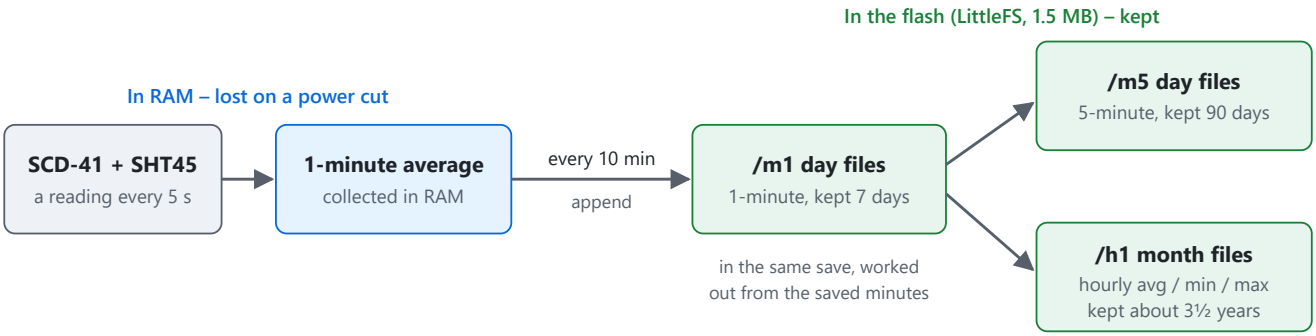


Figure 5 – How a reading becomes history. Only the part in RAM, at most 10 minutes, is lost on a power cut.

1. Every 5 s the SCD-41 delivers a CO_2 value. Together with the latest SHT45 values it is added to the average of the current minute, in RAM. Nothing is stored during the 30 s warm-up after a start.
2. Every 10 minutes (at :00, :10, :20 … plus 5 s) the finished minutes are appended to today's file in `/m1`.
3. In the same save, every finished 5-minute period and every finished hour is worked out from the saved 1-minute values and appended to `/m5` and `/h1`. Because they always come from saved data, a power cut doesn't leave a hole in them: the hour that was running is completed after the restart from what was saved before.
4. Files older than their retention are deleted. If less than 64 KB is free, the oldest hourly month is deleted.

Records are only ever added to the end of a file, never changed later. Flash memory can't change data in place, so editing old records would wear it out much faster.

7.3 Time

- The clock comes from the internet over Wi-Fi. All values are stored in **UTC**; the web page converts them to your local time, summer time included. That's why a day file starts at midnight UTC, not at local midnight.
- If the internet is down after a restart, the monitor doesn't know the time yet (it has no clock battery). It keeps up to **24 hours** of 1-minute values in RAM and gives them their correct times as soon as the clock is set. If the clock still isn't set after that, the oldest minutes are dropped.
- Without Wi-Fi details in `secrets.h` nothing is recorded, because the values would have no time.

7.4 Power cuts, moving and flash wear

- **Power cut:** the saved history is safe. LittleFS finishes a save completely or not at all, so files are never damaged. You lose at most the last 10 minutes. A firmware upload or the `RESET` button counts as a power cut too. Before you unplug or upload on purpose, type `save` in the Serial Monitor.
- **Moving the monitor:** the same as a power cut. The history stays on the device. In another home, only `secrets.h` needs the new Wi-Fi.
- **Flash wear:** flash wears when a 4 KB block is erased, and each block is rated for at least 100,000 erases. A save erases about 2–3 blocks, and LittleFS spreads the erases over the roughly 150 blocks that are reused again and again. That's about 2 erases per block per day, or around 100 years until the rating is reached.

7.5 What erases the history

> **Normal firmware uploads keep the history,** including uploads of the scanner and calibration sketches. It is lost only through:
>
> - a different **Flash Size** or **Partition Scheme** in the Tools menu (5.4). The firmware notices this at start, prints `History OFF: wrong partition scheme` and doesn't touch the flash. Set the menu back and upload again; the history is normally still there.
> - **Erase All Flash Before Sketch Upload: Enabled** in the Tools menu. Leave it Disabled.
>
> If you ever need either of them, save a CSV copy first (7.6).

7.6 Serial Monitor commands

Type a command in the Serial Monitor's input line and press `Enter`. The line-ending box next to it must not be set to "No Line Ending".

Command	What it shows or does
`status`	Wi-Fi, clock, flash use, minutes not saved yet, and the files of each level with their newest value.
`files`	Every history file with its size and number of values.
`csv m1 1`	The 1-minute values of the last day as CSV. `m5` = 5-minute, `h1` = hourly; the number is how many days back (`csv h1 400` = more than a year of hourly values).
`save`	Saves the minutes waiting in RAM now, for example before you unplug the monitor.
`sleep on`, `sleep off`	Switches sleep mode on or off (section 6). The answer is `Sleep mode on: LED off` or `Sleep mode off: LED on`.

Example answer to `status`:

```
Wi-Fi: connected to "MyNetwork", signal -58 dBm, web page at http://192.168.1.50/
Sleep mode: off
Clock: 2026-10-28 17:21:40 UTC
Flash: 32 KB of 1536 KB used (2%)
Not saved yet (in RAM): 2 minutes, next save in 505 s
1-minute values (m1): 1 file, 20261028.bin ... 20261028.bin, newest value 2026-10-28 17:19 UTC
5-minute values (m5): 1 file, 20261028.bin ... 20261028.bin, newest value 2026-10-28 17:15 UTC
Hourly values (h1): 0 files
```

And the start of `csv m1 1`:

```
time_utc,co2_ppm,temperature_c,humidity_pct,pressure_hpa
2026-10-28 17:05,612,22.41,45.30,
2026-10-28 17:06,618,22.43,45.12,
```

To keep a copy, select the lines in the Serial Monitor, paste them into a text file and save it as `.csv`. If Excel puts everything in one column, open it with *Data → From Text/CSV* and choose comma as the separator. Times are UTC. An empty field means the value wasn't measured. The web page's download (8.4) is easier: it uses local time and Excel's number format.

7.7 File format

Each file is a 16-byte header followed by fixed-size records in time order. Numbers are little-endian, and values are stored as whole numbers:

Field	Bytes	Unit	Means "missing"
Header (16 bytes): the letters `CO2H`, format version (1), minutes per record (1, 5 or 60), record size, creation time, 4 reserved bytes.			
1-minute and 5-minute record (12 bytes):			
time	4	UTC seconds since 1970, start of the minute or 5 minutes	–
CO_2	2	ppm	0
temperature	2 (signed)	0.01 °C	−32768
humidity	2	0.01 %RH	65535
air pressure	2	0.1 hPa	0
Hourly record (26 bytes): time (4), CO_2 average, minimum and maximum (3 × 2), the same for temperature and for humidity, average air pressure (2), number of minutes measured (1; 60 = complete hour), 1 reserved byte. Same units and "missing" values as above.			

7.8 How much fits

Part of the 1.5 MB	Space
1-minute files: 8 days of 17 KB (5 blocks of 4 KB each)	160 KB
5-minute files: 91 days of 3.5 KB (1 block each)	364 KB
LittleFS's own data, plus 64 KB that is always kept free	about 110 KB
Left for hourly values, at 5 blocks (20 KB) per month or 240 KB per year	**about 900 KB, roughly 3½ years**

LittleFS stores each file in whole 4 KB blocks, so space is counted in blocks. Keeping 5-minute values for 30 instead of 90 days (7.9) leaves room for about 4½ years of hourly values. A microSD card would allow much more.

7.9 Settings in `history.cpp`

Constant	Default	Meaning
`MINUTE_KEEP_DAYS`	7	Days of 1-minute values.
`FIVE_MINUTE_KEEP_DAYS`	90	Days of 5-minute values.
`SAVE_INTERVAL_S`	600	Seconds between saves, and the most a power cut can lose. Shorter means more flash wear.
`PENDING_MAX`	1440	Minutes kept in RAM while the clock isn't set (24 h, 17 KB of RAM).
`MIN_FREE_BYTES`	64 KB	When less is free, the oldest hourly month is deleted.

8 Web page

The monitor serves its own web page on your home network. It shows the live values, graphs of any stored day and a CSV download. It needs no app and no internet: any browser on a phone, tablet or PC in the same Wi-Fi works. The code is in `web.h` and `web.cpp`, the page in `web/index.html`.

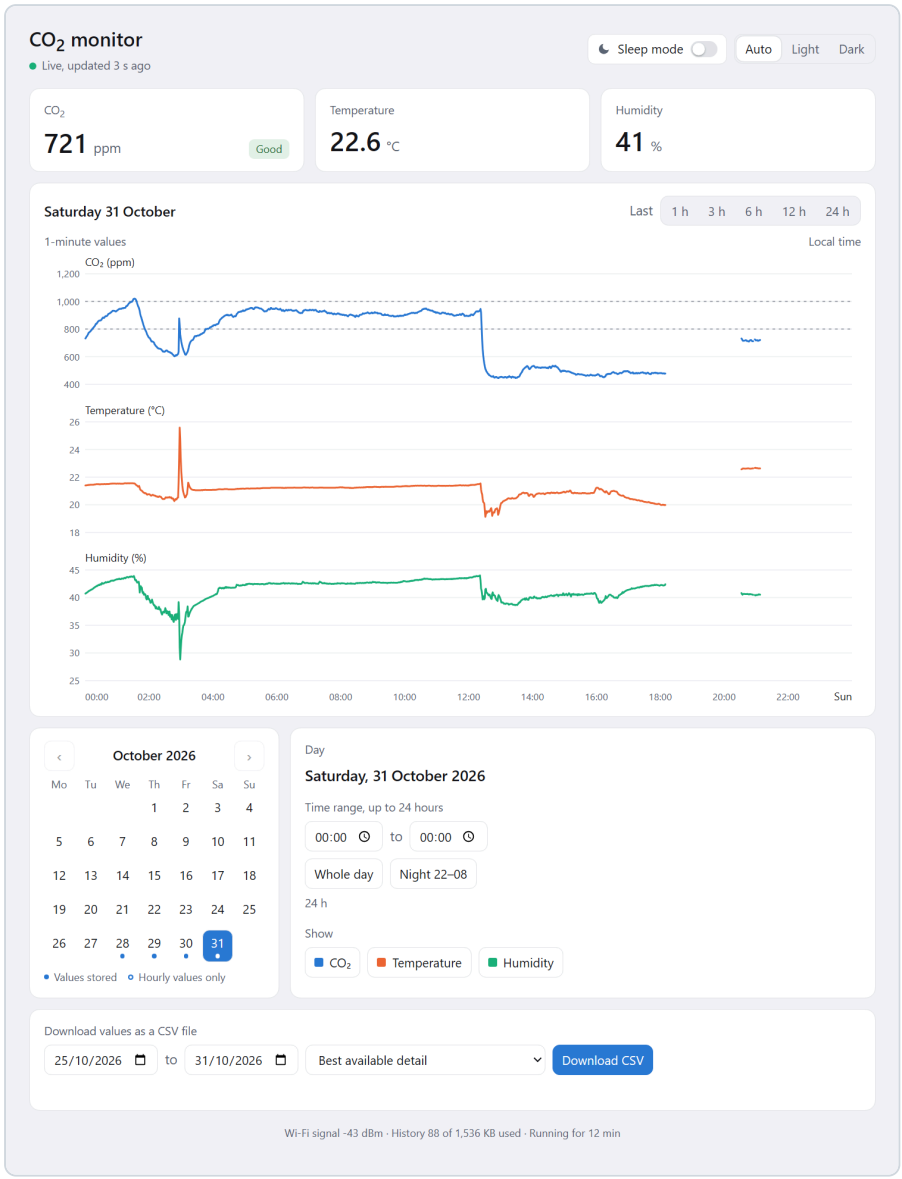


Figure 6 – The web page in light mode on 31 Oct 2026, with the *Sleep mode* switch (off) at the top right. The gap in the evening is a time with no values stored.

8.1 Opening the page

- At the start the Serial Monitor shows `Wi-Fi connected, web page at http://192.168.178.40/`, and `status` shows it too. Type that address into the browser and save it as a bookmark, or on the phone's home screen.
- The phone or PC must be in the **same Wi-Fi**. A guest network usually can't reach devices in the home network.
- There's no password. Anyone in your Wi-Fi can look at the values. The only thing the page can change is sleep mode; the history and the settings can't be changed from it.

8.2 Fixed address in the router

The router gives every device its address. It normally keeps it, but after a router reset or a long time switched off the monitor may get another one, and the bookmark stops working. Tell the router to always give the monitor the same address. The monitor is listed there as **co2-monitor**.

- **FRITZ!Box** (home addresses 192.168.178.x): *Heimnetz → Netzwerk → Netzwerkverbindungen*, pencil next to co2-monitor, tick *Diesem Netzwerkgerät immer die gleiche IPv4-Adresse zuweisen*, then OK. In English: *Home Network → Network → Network Connections*, *Always assign this network device the same IPv4 address*. The menu names differ a little between FRITZ!OS versions.
- **Other routers:** look for "DHCP reservation", "static lease" or "address reservation" in the LAN or DHCP settings.

> The address isn't fixed in the firmware on purpose. With a new router or in another network the monitor would then not get online at all, so it would get no time and store nothing. A name like `co2-monitor.local` isn't used either, because the Matter step brings its own name service and the two would clash.

8.3 What the page shows

Part	What it does
Status line	Green dot: a new reading arrived within the last 15 s (the page asks every 5 s). Red dot: no connection. "Waiting for the internet time" while the monitor's clock isn't set.
Live values	CO_2 with the same rating as the display (Good, OK, Ventilate, Bad air; limits in section 6), temperature and humidity. During the 30 s warm-up: "Warming up" with the seconds left.
Graphs	Right below the live values. One graph per value, all on the same time axis in local time; the heading above them names the time shown. Move the mouse over them, or drag a finger along them, to see the values at that moment. Dashed lines mark the CO_2 limits that fall inside the scale. A gap means nothing was measured (power off, restart, warm-up). While the range includes the current time, new values are added every minute.
Last 1–24 h	The buttons *1 h*, *3 h*, *6 h*, *12 h* and *24 h* above the graphs show the last hours up to now. The range then moves along with the time. Picking a day or changing a time ends it; a day picked after it is shown whole.
Detail	The last 7 days use the 1-minute values, up to 90 days the 5-minute values, older days the hourly averages with a shaded band from the hourly minimum to the maximum. The line above the graphs says which.
Day and calendar	Below the graphs: the chosen day and time range on the right, the calendar on the left (on a phone the day box comes first). In the calendar, a dot under a day means values are stored for it, a ring that only hourly values are left (days older than 90 days). Future days can't be chosen. The arrows change the month.
Time range	From and to, up to 24 hours. If "to" is earlier than "from", the range ends the next day. *Whole day* and *Night 22–08* set it with one click.
Show	Switches the CO_2, temperature and humidity graphs on and off. The browser remembers the choice.
Sleep mode	The switch at the top right, next to the theme buttons. Blue with the knob on the right = on: the monitor's RGB LED is off and its display shows a moon (section 6). It follows the monitor, so a long press on the button shows up here within 5 s. Greyed out while there is no connection.
Auto / Light / Dark	Auto follows the light or dark setting of the phone or PC. Light and Dark override it. The browser remembers the choice.
Bottom line	Wi-Fi signal, flash used by the history, and the time since the monitor started.

8.4 CSV download

Choose a start and an end date (both included) and the detail, then press *Download CSV*. The browser fetches the values from the monitor in parts and saves them as a file, for example `co2_2026-10-23_2026-10-29_1min.csv`.

- **Best available detail:** 1-minute values if the start date is within the last 7 days, 5-minute values within the last 90 days, otherwise hourly values. One file always has one level of detail.

- **Columns:** `time_local` (your time, summer time included), `time_utc`, then CO_2 in ppm, temperature in °C and humidity in %. Hourly files have the average, minimum and maximum of each, plus `minutes` (1-minute values in that hour; 60 = complete). Each time is the start of its minute, 5 minutes or hour. A pressure column appears once the BMP390 is added. An empty field means the value wasn't measured.
- **Excel:** the file uses the browser's number format. With German settings that's `22,5` with `;` between the columns, so Excel opens it straight into columns.

8.5 What the web page costs

Resource	Use
Program flash	About 60 KB more: 1,166,586 of 3,342,336 bytes (34%), up from 1,106,244. About 41 KB is the web server library and 12 KB the page (35 KB before compression). Plenty is left for Matter.
RAM	0.4 KB more fixed RAM, plus a few KB while a request is answered.
History storage	**Unchanged.** The page is part of the firmware, not of the 1.5 MB data partition, and reading the history doesn't wear the flash. The keeping times in 7.1 stay the same.
Measuring	Every answer is below 60 KB and takes well under a second. The main program waits meanwhile, but no reading is lost: the SCD-41 keeps its newest value until it is read.

8.6 Changing the page

- The page is `web/index.html`: HTML, CSS and JavaScript in one file, with no outside libraries. `web_page.h` holds a compressed copy that the firmware serves. After changing the page, run `web\make_web_page.ps1` in PowerShell, then upload the firmware.
- To try a change without uploading, open the file in Chrome or Edge with the monitor's address added: `…/web/index.html?device=192.168.178.40`. It then loads its data from the monitor.

The page gets its data from these addresses, which also work in your own scripts. Times are UTC seconds since 1970.

Address	Answer
`/api/now`	Live values, warm-up seconds, sleep mode, CO_2 limits, Wi-Fi signal and clock state (JSON).
`/api/info`	History on or off, flash use, minutes not saved yet, and the oldest file and days kept for each level (JSON).
`/api/days?from=…&to=…`	The UTC days with values and their finest detail, for the calendar marks (JSON, at most 45 days).
`/api/data?res=1|5|60&from=…&to=…`	The stored values as binary records, exactly as in the files (7.7). At most 2 days of 1-minute, 14 days of 5-minute or 92 days of hourly values per request.
`/api/sleep?on=1|0` (POST only)	Switches sleep mode on (1) or off (0) and answers `{"sleep":true}` or `{"sleep":false}`. Example from PowerShell: `Invoke-RestMethod -Method Post "http://192.168.178.40/api/sleep?on=1"`.

9 Testing

Test in this order. Each step builds on the one before, so when something fails you know where to look.

Step 1 – Power-on check

Plug in USB. The ESP32's power LED lights up, and so do the green power LEDs on the Adafruit boards (if present). Nothing gets warm. If anything gets hot, unplug immediately and check for a short between + and −.

Step 2 – I²C scan (`i2c_scanner.ino`)

Upload the scanner and open the Serial Monitor (115200). Every 5 s you should see:

```
Scanning I2C bus (SDA = GPIO23, SCL = GPIO22) ...
  0x3C  OLED display (SSD1306)
  0x44  SHT45 temperature / humidity
  0x62  SCD-41 CO2
3 device(s) found. Expected: 0x3C (or 0x3D), 0x44, 0x62
```

- **Nothing found:** check wires 3 and 4 (SDA/SCL swapped?) and the power wires 1 and 2.
- **Only one module missing:** check that module's four connections (OLED pin order! STEMMA cable fully seated?).
- **SHT45 missing but SCD-41 found:** check the STEMMA↔STEMMA cable between them.

Step 3 – Main program (`co2_monitor.ino`)

Set `SENSOR_ALTITUDE_M`, upload, open the Serial Monitor. Expected output:

```
CO2 monitor starting
SHT45 found
SCD-41 found, serial 0x...
SCD-41 automatic self-calibration (ASC): enabled
SCD-41 measuring
Warm-up: the first values follow in 30 s
Warm-up done, values are shown and saved from now on
CO2_ppm:612 Temperature_C:22.41 Humidity_pct:45.3
CO2_ppm:615 Temperature_C:22.42 Humidity_pct:45.2
```

The display shows "starting…", then "warming up" with a countdown from 30 s and `--.-` / `--%` at the bottom. After the countdown the temperature and humidity appear at once, and CO_2 within 5 s ("measuring…" until then). The RGB LED on the board goes off at the start and lights up dimly in the CO_2 colour with the first value. In normal room air it is green or yellowish. If the text is upside down, change `OLED_ROTATION` (2 ↔ 0) and upload again. **Serial Plotter** (Tools → Serial Plotter, 115200) draws the values as a graph. Untick Temperature and Humidity to see the CO_2 curve at a useful scale.

Step 4 – Button and display timeout

1. About 30 s after the warm-up ends (1 minute after start-up) the display goes dark. Serial output continues, so the sensors are still running.
2. Press the button. The display turns on immediately with current values.
3. Press again after ~20 s. The 30 s start over.

Step 5 – Sensor plausibility

- **Warm-up:** let everything run for **5–10 minutes** before judging the values. The SCD-41 needs a few minutes to settle, and the SHT45 needs time to reach room temperature after you handled it.
- **CO_2 breath test:** breathe gently toward the SCD-41 from 20–30 cm for a few seconds. Within 10–30 s the value jumps to well over 1000 ppm (several thousand is normal), then falls back over a few minutes. The LED goes through yellow and orange to red and back. This proves the sensor reacts. Don't breathe directly onto it from close up again and again; the moisture doesn't help it.
- **Humidity:** breathe near the SHT45. RH jumps by several percent and recovers within a minute.
- **Temperature:** hold a finger near (not on) the SHT45. Temperature rises slowly. Compare with another thermometer once it has settled; the SHT45 (±0.1 °C) is usually the more accurate one.

Step 6 – Fresh-air reference check

After the sensor has run for at least 10 minutes, take the setup **outdoors** (shade, not in direct wind, nobody breathing near it) on a power bank or laptop. Or air the room thoroughly: windows wide open, ideally a cross-draught, for 10–15 minutes. Outdoors it should read about **400–480 ppm**.

- Expected accuracy: ±(40 ppm + 5 % of reading), roughly ±60 ppm at 425 ppm. Readings between about 370 and 490 ppm outdoors are fine.
- A ventilated room rarely reaches true outdoor levels. 450–550 ppm indoors after airing is normal.
- Clearly off outdoors (e.g. above 550 or below 350 ppm): see Calibration (FRC).

Step 7 – Long run

Leave it running for at least 24 h, ideally a week. Typical pattern: CO_2 rises while people are in the room with the door and windows closed (a bedroom can reach 1500–2500 ppm overnight), and drops quickly when you air the room. If values drift or seem stuck, check the Serial output for errors.

Step 8 – Wi-Fi, clock and history

Enter your Wi-Fi in `secrets.h` (5.8), upload, and open the Serial Monitor. At the start you should see:

```
CO2 monitor starting
History: formatting the data partition (first start)
History: 8 KB of 1536 KB used
Wi-Fi: connecting to "MyNetwork"
SHT45 found
...
Wi-Fi connected, web page at http://192.168.1.50/
History: clock set to 2026-10-28 17:05:12 UTC
```

The "formatting" line appears only at the very first start, right after the first upload, so you'll usually miss it in the Serial Monitor. Every later start shows only the "KB used" line. Then:

1. Wait until the next full 10 minutes have passed (for example 17:10:05) and type `status`. `m1` and `m5` each show one file and a newest value.
2. Type `csv m1 1`. There is one line per minute since the start, and the CO_2 values match the live readings.
3. After the next full hour (plus 5 s), type `csv h1 1`. The first hour shows fewer than 60 in the `minutes` column, because the monitor started during that hour.
4. **Power cut test:** unplug the monitor for about 20 minutes, then plug it back in. In `csv m1 1` the gap starts at the last save before unplugging (at most 10 minutes earlier) and ends at the restart. Nothing else is missing.

5. **Optional, no internet at start:** switch off the router, restart the monitor, wait a few minutes, then switch the router back on. `status` shows "Clock: not set yet" until the router is back. Once the clock is set, the minutes measured without it appear in `csv m1 1` with the correct times.

Step 9 – Web page

1. Open the address from the Serial Monitor (`status` shows it too) in a browser on a phone or PC in the same Wi-Fi. The live values appear within 5 s, and the status line says "Live".
2. The calendar shows a dot under every day with values. Pick one, and the graph shows that day. Move the mouse over the graph: the values match `csv m1 1` for that minute. The page shows local time, the Serial Monitor UTC.
3. Press *3 h* above the graphs: they show the last 3 hours, and a minute later the range has moved on by a minute. Press *Night 22–08*: the range ends the next morning. Switch off two of the values under *Show*, and only one graph is left.
4. Download the last 7 days with *Best available detail* and open the file in Excel. There is one row per minute, already split into columns.
5. Switch your phone or PC between light and dark mode. With *Auto* the page follows.

Step 10 – Sleep mode

1. Hold the button. The display wakes at once; after 2 s it shows the big moon with "Sleep mode on - LED off", and the RGB LED goes out. Let go. After 2.5 s the display goes dark.
2. Press the button briefly. The values appear with a small moon at the top, and the display goes dark again after 30 s. The LED stays off.
3. Open the web page: the *Sleep mode* switch is on. Switch it off. The LED lights up in the CO_2 colour within a moment, and the Serial Monitor shows `Sleep mode off: LED on`.
4. Switch it on again on the page, then press RESET on the board. After the restart the Serial Monitor shows `Sleep mode on: LED off`, and the LED stays off after the warm-up. `csv m1 1` still gets a line every minute: sleep mode doesn't stop the recording.

Tested on the board on 31 Oct 2026: web switch, `/api/sleep`, the Serial commands and the restart behaved as above.

10 CO_2 calibration (SCD-41)

The SCD-41 is **calibrated at the factory** by Sensirion and works out of the box. Over time, mechanical stress, soldering, shipping and ageing can shift its zero point (the "offset"). Two mechanisms correct this:

	ASC – automatic self-calibration	FRC – forced recalibration
Who does it	The **sensor itself** (firmware inside the SCD-41)	**Your code** sends one command
Default	**On** from the factory	Only when you trigger it
Code needed	None. Our main sketch leaves it as it is.	The optional `scd41_calibration.ino` tool
How it works	Watches the readings over days. Assumes the **lowest** level seen in each period was fresh outdoor air (target 400 ppm) and slowly corrects the offset. First correction after ~**44 h** of operation, then every ~**156 h** (6.5 days).	You put the sensor in air of known CO_2 (outdoors ≈ 425 ppm), let it run ≥ 3 min, and tell it "this is 425 ppm". The correction is applied immediately.
Requirement	The sensor sees fresh air (~400 ppm) **at least once per week of operating time**, while running **≥ 4 hours in one go**.	Stable, known reference air, and the sensor running in its normal mode for ≥ 3 minutes beforehand.

Use when	Normal homes and offices that get aired regularly.	Readings are clearly off in fresh air, or the room never gets fresh air (then also switch ASC off).

Your questions, answered

Is the CO_2 sensor calibrated by the code or by the sensor itself?

Both **Normally the sensor does it itself.** ASC runs inside the SCD-41 and is switched on at the factory, so no code is needed. Our main program doesn't touch the calibration; it only sends the altitude at every start. **FRC** is triggered by code when you know the true CO_2 level. The separate `scd41_calibration.ino` sketch does exactly that.

How should I start testing – open the door and let fresh air in, or just plug it in?

Just plug it in No fresh air is needed to start. Give it 5–10 minutes, do the breath test (CO_2 must jump), then do the fresh-air check from Testing, step 6. That check tells you whether the factory calibration is still good; it doesn't calibrate anything by itself.

Do I need fresh air every time I reconnect the board?

No The SCD-41 stores its calibration history (ASC and FRC) in its **own EEPROM automatically**, so it survives unplugging. You only need fresh air regularly *for ASC to keep working long term*: at least once per week of operating time, while the sensor runs for 4+ hours at a stretch. Normal airing of the room does that. Short sessions (unplugged after an hour) don't advance ASC, but they don't lose anything either.

What is stored where

Item	Stored	After unplugging
Factory calibration, ASC history, FRC correction	Sensor EEPROM (automatically)	**Kept**
Altitude, ASC on/off, temperature offset, ASC target	Sensor RAM; EEPROM only after `persistSettings()`	Lost, back to the saved values. Our code re-sends the altitude at every start, so there's no need to persist it.

> The EEPROM is rated for only ~2000 writes. **Never call `persistSettings()` in `loop()`**, only once after a deliberate settings change. Our sketches never call it.

When ASC can go wrong

- **The room never gets fresh air** (always occupied, never aired). ASC then treats e.g. 700 ppm as "400" and pulls all readings too low. Switch ASC off with `setAutomaticSelfCalibrationEnabled(0)` + `persistSettings()` and use FRC every few months instead.
- **Target 400 vs real outdoor air:** ASC assumes 400 ppm, but outdoor air today is ~420–430 ppm. ASC can therefore read ~25 ppm low. That is well inside the sensor's accuracy, so it's fine for step 1. It could later be changed with `setAutomaticSelfCalibrationTarget()`.

FRC procedure (only if needed)

1. Upload `firmware/scd41_calibration/scd41_calibration.ino` (set `SENSOR_ALTITUDE_M` the same as in the main sketch).

2. Take the complete setup **outdoors** on a power bank or laptop: shade, away from roads, chimneys and people. Don't do it at an open window indoors; indoor air near a window is often 450–600 ppm.
3. Wait. The display counts down 5 minutes and shows the CO_2 value and the **spread** (max − min) of the last minute. A spread under ~20 ppm means stable air.
4. When it shows "press button", press the button (or send `c` in the Serial Monitor).
5. The display shows "Done! corr ±xx ppm". The new reading should now be close to 425 ppm. The correction is stored in the sensor automatically.
6. Upload `co2_monitor.ino` again.

> The target is `FRC_TARGET_PPM = 425` (global outdoor background in 2026 is about 420–430 ppm). A wrong reference makes the sensor worse, not better, so only run FRC in genuinely fresh air.

11 Hardware notes

ESP32-C6-DevKitC-1-N8

- **Chip:** ESP32-C6 (in the ESP32-C6-WROOM-1-N8 module). 32-bit RISC-V main core up to 160 MHz plus a 20 MHz low-power core, 512 KB SRAM, **8 MB flash**.
- **Radio:** Wi-Fi 6 (2.4 GHz only), Bluetooth 5 LE, IEEE 802.15.4 (Zigbee / Thread / Matter). Not used yet. Keep the antenna end free of metal.
- **3.3 V logic, GPIOs not 5 V tolerant.** The 5V pin carries USB voltage. Never connect it to a GPIO or to the sensors' bus.
- **Two USB-C ports:** "USB" = native USB (Serial/JTAG, GPIO12/13), "UART" = USB-to-serial bridge (GPIO16/17). Either one powers and programs the board.
- **On board:** BOOT button (GPIO9), RESET button, addressable RGB LED (WS2812B on GPIO8), red power LED ("3.3 V Power On LED", always on, see below). Jumper **J5** (2-pin header next to the "3V3" marking) carries the 3.3 V supply to the ESP32 module. It is fitted at the factory and **must stay fitted**; it is only removed to measure the module's current with an ammeter. Without it, the USB chip still shows up on the PC, but the ESP32 itself has no power.
- **Strapping pins** GPIO4, 5, 8, 9, 15: don't connect buttons or pull-downs to them.
- **Breadboard fit:** 25.4 × 51.8 mm, pin rows 22.86 mm (9 holes) apart. In columns b and i it leaves one free hole per pin on each side. The PCB extends ~12 mm past pin 16 (USB end) and the module overhangs ~8 mm past pin 1 (antenna end). With pins in rows 3–18, the USB ports stick out about 2.5 mm past the breadboard (in rows 4–19 they would end flush with it, behind its joining nubs) and the antenna covers rows 19–21, which leaves rows 22–30 free for the button and the bus.

Waveshare ESP32-C6-DEV-KIT-N16 (for the case, not bought yet)

- **Same pins:** two rows of 16 with the same pins in the same order as the DevKitC-1 (J1/J3), the same ESP32-C6-WROOM-1 module family, the same SGM2212 3.3 V regulator (800 mA) and the RGB LED (WS2812B) on GPIO8. The firmware's pins don't change.
- **One USB-C port** instead of two: a USB hub chip (CH334F) passes both the native USB and a USB-to-serial chip (CH343P) through the one port, so the PC shows two COM ports. Windows may need WCH's CH343 driver.
- **16 MB flash.** The present board settings (8 MB, `default_8MB`) work on it unchanged. The stored history stays on the old board: download the CSV first (8.4).
- **Size and holes:** 55 × 25.36 mm, first pin 12.0 mm from the USB end, 4 corner holes ⌀2 at 1.77 mm from both edges; the DevKitC-1 has none. In the case it lies upside down on 4 posts (section 13).

- **Red power LED** (2 kΩ, no jumper), brighter than the DevKitC-1's. In the case it faces the floor, next to the floor vents: tape over it for the night.
- Sold by Berrybase with the headers soldered on (WS-30547, €9.90). Waveshare is a long-standing Shenzhen maker; its board is a clean copy of Espressif's design, but its wiki is partly out of date.

Adafruit SCD-41 (Sensirion SCD41)

- **Principle:** photoacoustic NDIR. It measures real CO_2, unlike cheap "eCO_2" gas sensors that only estimate it.
- **Range / accuracy:** 400–5000 ppm specified at ±(40 ppm + 5 % of reading); outputs up to 40 000 ppm. (The SCD-40 is only specified to 2000 ppm.)
- **Modes:** periodic (every 5 s, ~17 mA; used here), low-power periodic (every 30 s), single shot (SCD-41 only, ~0.5 mA at one reading per 5 min; useful for battery operation later).
- **I²C address 0x62** (fixed). 3.3–5 V supply; the breakout has a regulator and level shifting. A clean, stable supply matters for accuracy.
- **Built-in T/RH:** the SCD-41 always measures temperature and humidity; it needs them internally to compensate the CO_2 value, and this cannot be switched off. Every reading returns all three values. The code reads them and simply ignores T/RH, because the sensor warms itself slightly and the SHT45 is more accurate. Ignoring them has **no effect on CO_2 accuracy**, and neither does its temperature offset setting (it only shifts the reported T/RH). The SCD-41 cannot take temperature or humidity from the SHT45.
- **Air pressure:** this *does* affect CO_2 accuracy, so set your altitude in the code. Weather changes air pressure by a few percent; a pressure sensor added later could feed live values with `setAmbientPressure()`, which works during periodic measurement.
- **Handling:** the weak spot is the **white membrane on top of the metal cap**. It keeps dust and water out and lets CO_2 in. Sensirion: don't remove, damage, imprint or cover it. So don't touch or press it, and don't tape over it, clean it, or blow compressed air at it. The sensor is an optical system: avoid strong force on the cap, especially sideways, and drops onto hard floors. ESD rating is only 2 kV (HBM), so touch something metal first. Chemical fumes don't affect the CO_2 measurement the way they affect the SHT45's humidity reading. Keep dust, direct sunlight and strong drafts off it. Leave space for air to reach it. If it ever reads clearly wrong in fresh air after a knock, a forced recalibration fixes the offset (section 10). Calibration persists in its EEPROM.

Sensor choice: SCD-41 vs SCD-30

	SCD-41 (used)	SCD-30
Principle	Photoacoustic NDIR	Dual-channel NDIR (reference channel for long-term stability)
Specified range	400–5000 ppm	400–10 000 ppm
Accuracy	±(40 ppm + 5 %) → ±90 ppm at 1000 ppm	±(30 ppm + 3 %) → ±60 ppm at 1000 ppm
Sensor size	10.1 × 10.1 × 6.5 mm	35 × 23 × 7 mm
Power	~17 mA (every 5 s); single-shot mode ~0.5 mA (every 5 min)	~19 mA (every 2 s); no low-power single-shot mode
I²C	Standard, up to 400 kHz	Max 100 kHz, needs clock stretching (up to 150 ms), a known source of trouble with some microcontrollers
Self-calibration	Fresh air once per week of operation, runs of ≥ 4 h	7 days continuous operation, ≥ 1 h fresh air per day
Adafruit breakout	STEMMA QT, 0x62	STEMMA QT, 0x61; $58.95 and often out of stock

Conclusion: for a room monitor the SCD-41 is the better fit. It is small, has trouble-free I²C, and offers a low-power mode for a later battery version, and indoor air stays well inside 400–5000 ppm. The SCD-30 wins on paper accuracy, range and long-term stability. Choose it for mains-powered builds that may exceed 5000 ppm (greenhouse, grow tent, fermentation room) or where the best absolute accuracy matters more than size and power.

Adafruit SHT45 (Sensirion SHT45)

- **Accuracy:** ±0.1 °C typical (5–60 °C), ±1.0 %RH typical (20–70 %RH). Range −40…125 °C, 0…100 %RH.
- **I²C address 0x44** (fixed). 3–5 V supply; the breakout has a regulator, level shifting and pull-ups.
- **Built-in heater:** for drying the sensor after condensation. While it runs, temperature readings are useless, so the code keeps it **off**.
- **Placement is everything:** the ESP32, the SCD-41, the display and your hand all give off heat. Mount the SHT45 at the end of the cable, away from warm parts and out of direct sunlight. Airflow improves response time.

Sensor handling: what can harm them

Source	Effect	What to do
Cleaning sprays (ammonia, bleach, alcohol), solvents (acetone / nail polish remover, paint, varnish), glue, silicone or epoxy while curing	**Permanent damage possible.** The SHT45's humidity reading drifts, in most cases for good. Sensirion names acetone, ethanol, isopropanol, toluene, ammonia and acids.	Don't spray or wipe near the sensors. Take them out of the room while painting or gluing, and air the room until the smell is gone.
Aerosols (hairspray, deodorant, spray paint), cooking grease, smoke	Coat the sensor openings.	Not in the kitchen or bathroom, no spraying nearby.
Water: spills, condensation (bathroom steam, cold window sill)	A short circuit can kill the boards. The SCD-41 is only rated for non-condensing air.	Keep everything dry. Unplug at once after a spill.
Static electricity	Can destroy the chips.	Touch something metal first (e.g. a radiator), hold the boards by their edges.
Pressing, dropping, pulling on cables	Broken connectors or solder joints.	Don't press on or poke into the SCD-41's sensor. Pull STEMMA plugs by the plug.
Vibration, loud bass (on top of a subwoofer or speaker)	No damage. Can make the SCD-41's readings noisier; Sensirion recommends keeping it decoupled from vibration.	Put it on a shelf or desk, not on the speaker.
Magnets (speakers, subwoofers, monitors, phone cases)	**No effect.** Neither sensor uses or reacts to magnetic fields.	Nothing.
Heat: sun, radiator, monitor, laptop vent, the ESP32, your hand	No damage at room conditions, but temperature reads too high and humidity too low.	See placement below.

How likely is it? Sensirion's rule of thumb: air that people can breathe for a long time without harm won't contaminate the sensor. Normal cleaning, cooking or a candle across the room is fine. The real risks are high concentration and long exposure: the sensor right next to curing glue or silicone, painting in a closed room, wiping the sensor with alcohol, or storing it in a box with solvents. Chemicals only affect the **humidity** reading; temperature isn't affected.

If the SHT45 does get contaminated

1. **Check it with the salt test.** Put a spoonful of table salt with a few drops of water (a damp paste, not dissolved) in a bottle cap. Put the cap and the SHT45 board in an airtight container or a zip bag, and run the STEMMA cable

out through the seal. Make sure the salt doesn't touch the board. After 8–12 h at a steady room temperature the air inside is at **75 %RH**. The SHT45 should read 74–77 %RH. If it does, it's fine.

2. **Mild drift:** air the room well and leave the sensor running for a few days. Part of the contamination outgasses by itself.
3. **Heat it with its built-in heater.** In Sensirion's test, SHT4x sensors were exposed to saturated acetone for 20 h, far worse than anything at home. Heating with the on-chip heater (110 mW, 80 min, about 110 °C) brought them fully back within specification. The heater runs for at most 1 s per command and must stay below a 10 % duty cycle. On the Adafruit board the thicker PCB carries heat away, so the sensor gets less hot and the treatment has to run longer. Then let it rehydrate in the salt container (12 h at 75 %RH, the second half of Sensirion's reconditioning procedure).
4. **Oven baking** (Sensirion's full reconditioning: 10 h at 100–105 °C, then 12 h at 75 %RH) is **not recommended** for the breakout: the STEMMA QT connectors are only rated up to 85 °C, and a kitchen oven can't hold 100 °C accurately.
5. **Last resort:** correct a known, stable offset in software, or replace the board. It plugs in with the STEMMA cable, with no soldering.

Touching is fine if you hold the boards by their edges after touching something metal. A finger on the SHT45 shows up as a temperature and humidity rise that disappears within a minute. Don't put fingers on the sensor openings, because skin oil stays there.

Power LEDs: taping over them or switching them off

Three LEDs stay lit in sleep mode, because they are wired straight to the 3.3 V supply and not to a GPIO. No firmware can switch them off.

LED	Wiring	How to get it dark
ESP32-C6-DevKitC-1, red	D5 with R11 (5.1 kΩ) on the 3.3 V rail; no jumper. Pulling J5 doesn't help: the LED is on the board side of J5, and the ESP32 would lose power.	Tape over it; no sensor is near it, so any tape is fine there. For good: unsolder R11 (the board then shows no sign of power).
SCD-41, green "on"	D1 with R1 (10 kΩ) through jumper SJ3, front side.	**Best:** cut the thin trace between the two pads of SJ3 (2 in Figure 7) with a craft knife. A blob of solder across the pads undoes it. Or tape, within the green area.
SHT45, green "on"	D1 with R1 (10 kΩ); no jumper on this board.	Tape, as small as possible (Figure 7). For good: unsolder R1 (0603 size, next to the LED).

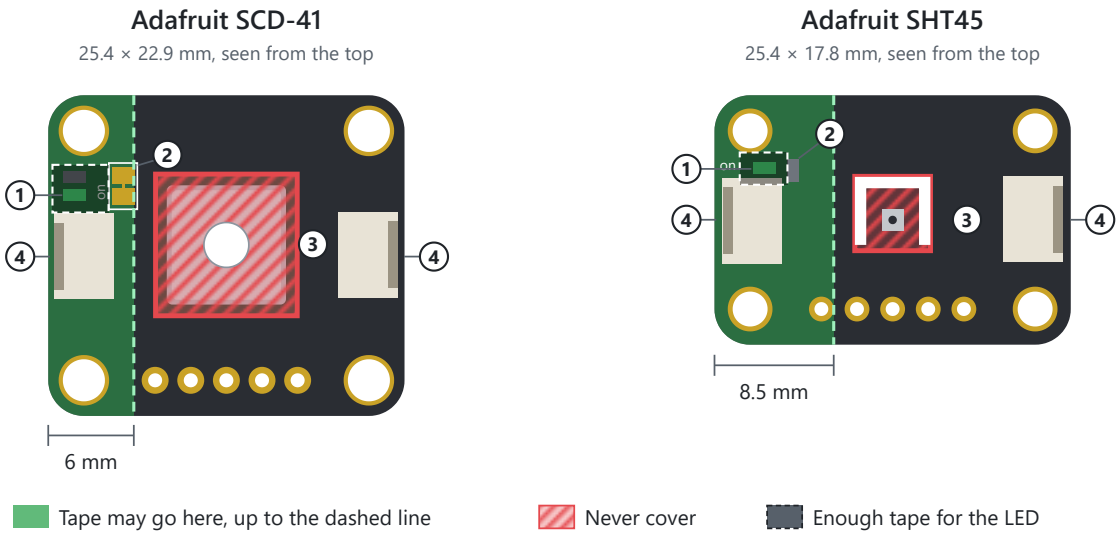


Figure 7 – Where electrical tape may go, to scale from Adafruit's board files (top view, pin header at the bottom, "on" LED top left). (1) Green power LED. (2) SCD-41: jumper SJ3, cut it to switch the LED off for good; SHT45: resistor R1, the only way to switch its LED off for good. (3) The sensor: on the SCD-41 the top of the metal cap with its white membrane, on the SHT45 the tiny chip with its 0.6 mm opening and the U-shaped slot cut around it. (4) Plug openings: don't fold tape over the board edge here. The dashed line ends the green area 1.5 mm before the sensor: 6 mm from the left edge on the SCD-41, 8.5 mm on the SHT45. Check against your own boards; the LEDs sit at 1–3 mm (SCD-41) and 2.5–4.5 mm (SHT45) from the left edge.

- **The hard limit** is the sensor itself. Tape on the SCD-41's cap blocks the membrane that lets CO_2 in (Sensirion: don't remove, damage, imprint or *obstruct* it). Tape on the SHT45's chip covers its humidity opening.
- **Why keep it small anyway:** PVC electrical tape and its glue give off vapours, and humidity sensors pick them up (see *Sensor handling* above). The SHT45 and the humidity sensor inside the SCD-41 are both affected. Sensirion's handling instructions name only a few tapes as safe near their humidity sensors (for example 3M Kapton tape 92); ordinary black electrical tape isn't one of them. A piece of about 4 × 3 mm in an open room is a small risk; a big piece inside the closed sensor chamber of the case is a bigger one.
- **Allowed but pointless:** tape on the rest of the board (unshaded) does no harm, as long as it keeps off the sensor, the plug openings and any mounting hole you want to screw through. It just adds more glue.
- For the finished case: cut SJ3 on the SCD-41 and, if you can solder, unsolder R1 on the SHT45. Then no tape is needed in the sensor chamber.

Sensor placement

- In the room where you spend the most time (bedroom or office), at about **desk to head height (0.8–1.5 m)**.
- **About 1 m away from people's faces.** Directly in front of you on the desk, every breath reaches the sensor and the CO_2 value jumps around.
- Not near windows, doors, air vents, radiators, the stove, a humidifier, or in direct sunlight.
- Free air on all sides: not in a drawer, a closed shelf or behind the monitor.
- The SHT45 beside or below the ESP32, never above it (hot air rises), and at least 5–10 cm from the ESP32, the SCD-41 and the monitor.
- Air the room now and then: the SCD-41's self-calibration needs fresh-air levels at least once a week (section 10).

Joy-it 0.96" OLED (SBC-OLED01)

- 128×64 monochrome OLED, **SSD1306** controller, I^2C. Address **0x3C** as delivered; **0x3D** after moving the resistor on the back.
- 3.3–5 V supply, 3.3 V logic. Pin order on this module: **GND, VCC, SCL, SDA** from left to right, header at the top edge. With the pins pointing out of the back, it lies flat when plugged into the breadboard.

- OLEDs **burn in** like OLED TVs and phones, but without their protection features: pixels that are lit for thousands of hours age faster and leave a faint ghost image. The fixed parts of the layout (the "CO2 ppm" label, the line, the °C sign) would show first. The 30 s timeout keeps the total on-time to minutes per day. If the display is ever kept on permanently, lower `OLED_BRIGHTNESS`, shift the image by a pixel every few minutes, or switch it off at night.
- **Brightness:** the SSD1306 contrast setting (command 0x81, 0–255) changes brightness only roughly 2× between 0 and 255. Darker than that needs extra settings (pre-charge period and V_{COMH} level).
- "Display off" (SSD1306 command 0xAE) turns the panel off and keeps its content; the controller then draws only a few µA.

Push button (4 legs)

- The four legs form two internally connected pairs; pressing the button joins the pairs. Using two **diagonally opposite** legs always works.
- No resistor is needed: the code enables the ESP32's internal pull-up (`INPUT_PULLUP`). Released = HIGH, pressed = LOW. The code debounces it (30 ms).

STEMMA QT cables

- 4-pin JST SH, 1.0 mm pitch, same as SparkFun Qwiic. Colours: **black = GND, red = V+, blue = SDA, yellow = SCL**.
- Each Adafruit board has two connectors, so modules can be chained. The connectors are small; pull them out by the plug, not the wires.

400-point breadboard

- 30 rows × 2 × 5 holes plus 4 power rails. Holes a–e of a row are connected, f–j are connected, not across the gap.
- Some breadboards have rails that are split in the middle. Check with a multimeter, or bridge the halves if yours is like that.

12 Troubleshooting

Symptom	Cause / fix
No COM port appears	Charge-only cable → use a data cable. Missing driver for the "UART" port → see the next row. Try the other USB-C port.
"CP2102N USB to UART Bridge Controller" under *Other devices*, Code 28	The CP210x driver isn't installed → install the Silicon Labs CP210x Universal Windows Driver (5.5). Or use the "USB" port with *USB CDC On Boot* = *Enabled*, which needs no driver.
Upload: "Failed to connect to ESP32-C6"	Hold BOOT, tap RESET, release BOOT, upload again. Select the right port. Try Upload Speed 115200. Check that the J5 jumper is fitted.
Serial Monitor shows nothing	Baud must be 115200. "USB CDC On Boot" must match the port (UART port → Disabled, USB port → Enabled). Press RESET.
Serial Monitor shows garbage	Wrong baud rate.
Scanner finds no devices	SDA/SCL swapped (GPIO23 = SDA, GPIO22 = SCL). No 3.3 V or GND on the bus rows (wires 1 and 2). OLED or STEMMA pins in the wrong rows.
Display found but stays dark	It timed out → press the button. Check `OLED_ADDRESS` (0x3C vs 0x3D, see scanner output).
RGB LED stays off although CO_2 values come in	Sleep mode is on: the display shows a moon at the top, `status` says "Sleep mode: on". Hold the button for 2 s, or use the switch on the web page.

Holding the button doesn't switch sleep mode	Hold it for a full 2 s, until the big moon or sun appears. A shorter press only wakes the display.
Red LED on the ESP32 and green LEDs on the sensors stay on in sleep mode	They are power LEDs on the supply, not under software control. Cover them or disconnect them (section 11, Figure 7).
Display text is upside down	Change `OLED_ROTATION` in the sketch (2 ↔ 0).
Display shows noise or a picture shifted by 2 pixels	The module has an SH1106 controller, not an SSD1306 (common on 1.3" modules). Switch the code to the *Adafruit SH110X* library.
Button does nothing	Button not across the gap, or both wires on the same internal leg pair → use diagonal legs f25 / e27. Check wire 5 (j7 = next to GPIO2).
"measuring…" never goes away after the warm-up	SCD-41 is not delivering values; check Serial for "SCD-41 not found" and the STEMMA cable.
CO_2 fluctuates by 10–30 ppm	Normal. People nearby, drafts and doors cause real changes too.
Temperature reads 1–3 °C too high	SHT45 too close to the ESP32, SCD-41, display, or in sunlight. Move it away.
Outdoor CO_2 clearly not ~420 ppm	Let ASC work for a week with regular airing, or run the FRC tool outdoors.
Board doesn't start after adding hardware	Something is connected to a strapping pin (GPIO4/5/8/9/15). Move it to a free GPIO.
Compile error about `SensirionI2cScd4x`	Old SCD4x library (0.x) installed → update to 1.x in the Library Manager.
Compile error "secrets.h is missing"	Copy `secrets_example.h` to `secrets.h` and enter your Wi-Fi (5.8).
Serial: `Wi-Fi: enter your network in secrets.h`	`secrets.h` still has the example name. Enter your network and upload again. Without Wi-Fi there's no clock, so nothing is recorded.
"Wi-Fi: connecting…" but never "connected"	Wrong name or password (both are case-sensitive), a 5 GHz-only network, or the router is too far away.
`status` : "Clock: not set yet" although Wi-Fi is connected	The router has no internet or blocks time requests (UDP port 123). The values wait in RAM for up to 24 h. If your router offers time itself, enter its address as `NTP_SERVER_1` .
Serial: `History OFF: wrong partition scheme`	Tools menu: Flash Size **8MB**, Partition Scheme **8M with spiffs (3MB APP/1.5MB SPIFFS)** (5.4), then upload again. The history is normally still there.
Commands in the Serial Monitor do nothing	The line-ending box next to the input line is set to "No Line Ending". Choose "New Line".
Web page doesn't open	The phone or PC is in another network (guest Wi-Fi, mobile data). Or the address has changed: `status` shows the current one, and a fixed address in the router (8.2) stops it changing.
Web page: "No connection to the monitor"	The monitor is unplugged, restarting or out of Wi-Fi range. The page keeps trying every 5 s and continues by itself.
Web page: "waiting for the internet time"	Same cause as "Clock: not set yet" above. The graphs appear as soon as the clock is set.

13 Case (design v4)

> **Design v4, 3–4 Nov 2026.** A compact case without the breadboard: **86.3 × 72 × 40 mm** instead of 105 × 86 × 40 mm, with the same look as v3 (rounded box, 4 mm side slots, display and light-sensor windows in the front, the button with the LED below it, a speaker grille and 4 screws in the lid, rubber feet). It is built for the **Waveshare ESP32-C6-DEV-KIT-N16**, which is still to be bought; until then the breadboard prototype of sections 2–12 stays as it is. Figures 8–11 are pictures of the 3D model (*3D model* below). The v3 design, with the breadboard in the case, stays in `cad/case.py` for reference.

The design

- **Size:** 86.3 × 72 × 40 mm, 43.5 mm high with the rubber feet: about 30 % less volume than v3. The length comes from the two boards on the column-j wall plus the button behind the front; the height from the display with the light sensor under it, and from the ESP32 with the speaker above it.
- **Sensor chamber along the column-a side**, behind the double wall with its air gap, as in v3. It now holds only the **SCD-41** (back) and the **SHT45** (front, farthest from the ESP32), 8.5 mm apart, both screwed to standoffs through the side slots.
- **ESP32 without a breadboard:** the Waveshare board lies **upside down** on 4 posts at the back, its pins pointing up, so the jumper-wire plugs go straight onto them. Its USB-C port sits low in the back wall. Three corners are screwed (M2); the fourth only rests on its post, because the module sits right next to that hole.
- **Column-j compartment:** a **wall** separates a small compartment along the column-j side from the ESP32. It has its own 9 side slot columns. The **BMP580** stands in its front half and the **SGP40** air-quality sensor (for later) in its back half. Both face the wall and are screwed on from outside through the slots, like the chamber sensors. They get room air straight through the slots, away from the ESP32's warmth.
- **Vents:** the sensor chamber breathes through its side slots, 2 slot columns in its back end and 7 floor vents between the feet. The column-j compartment has its 9 side slot columns. The ESP32 takes cool air in through 6 floor vents under the board, and its warm air leaves through 5 slots high in the back wall, around the USB port. The front and the lid have no vents apart from the speaker grille. No fan.
- **Front face:** the OLED behind its window, its glass 1 mm behind the outside (holder as in v3, 1.6 mm higher). The **light sensor** (right-angle VEML7700) lies under the display on 4 low posts and looks out through a window shaped like its sensor. The **panel button** (Berrybase 10130) sits level with the middle of the display on the column-j side, held by its own nut. The **8 mm LED** (PL9823-F8) sits below it in a sleeve, its dome 1.5 mm out of the front, on the far side from the light sensor.
- **Speaker** under the lid over the ESP32, with the grille above it. Its plug faces the double wall, right above the ESP32's 5V, GPIO3 and GND pins.

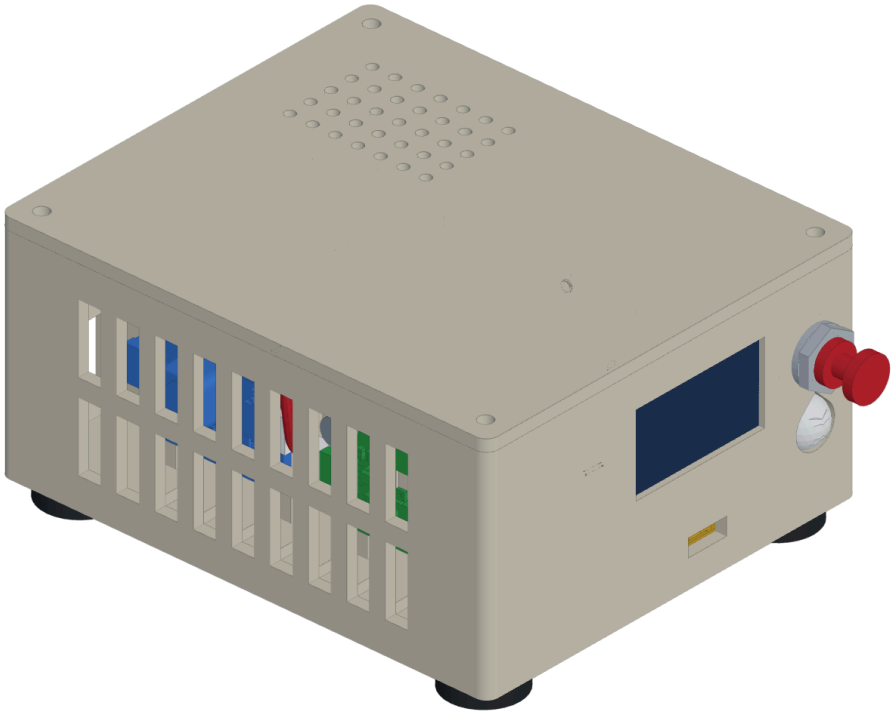


Figure 8 – The closed case from the front and the sensor-chamber side (picture of the 3D model). Front: display window, the light-sensor window under it, the panel button with its nut and the LED below it. Side: the chamber's slots with the SCD-41 (blue) and SHT45 (green) behind them. Lid: speaker grille and 4 screws.

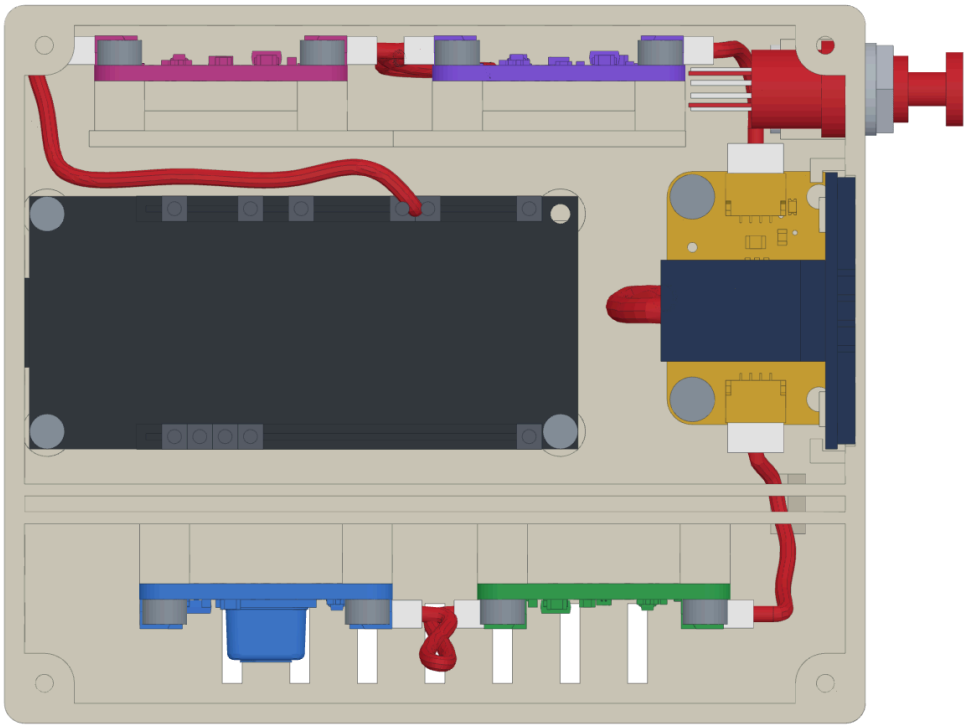


Figure 9 – Top view with the lid and speaker removed: back (USB) on the left, front on the right, sensor chamber at the bottom. Middle: the ESP32 upside down on its 4 posts, with jumper plugs on the pins it uses (grey). Top: the column-j compartment behind its wall, with the SGP40 (pink, later) and the BMP580 (purple). Right: light sensor (yellow) under the display (dark blue), and the button (red). Bottom: SCD-41 (blue) and SHT45 (green). Red: STEMMA QT cables and the display's wires.

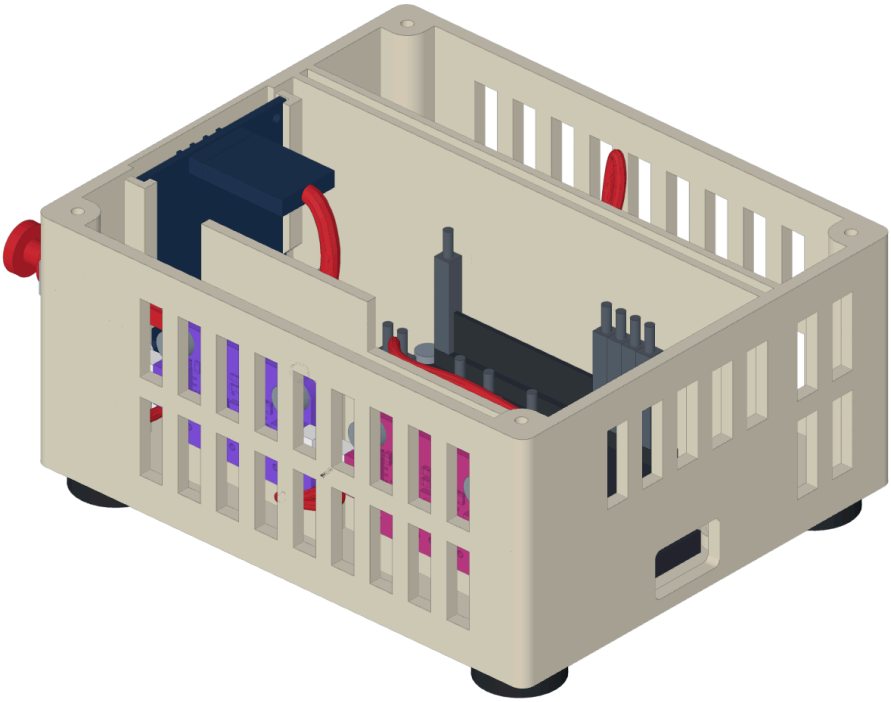


Figure 10 – From the back and the column-j side, lid and speaker removed. The wall between the ESP32 and the compartment stays 0.5 mm under the speaker and reaches the lid in front of it. The cable from the ESP32 runs along the wall and round its back end to the SGP40 (until then, on to the BMP580). In the back wall: the USB opening and the 5 vent slots above it.

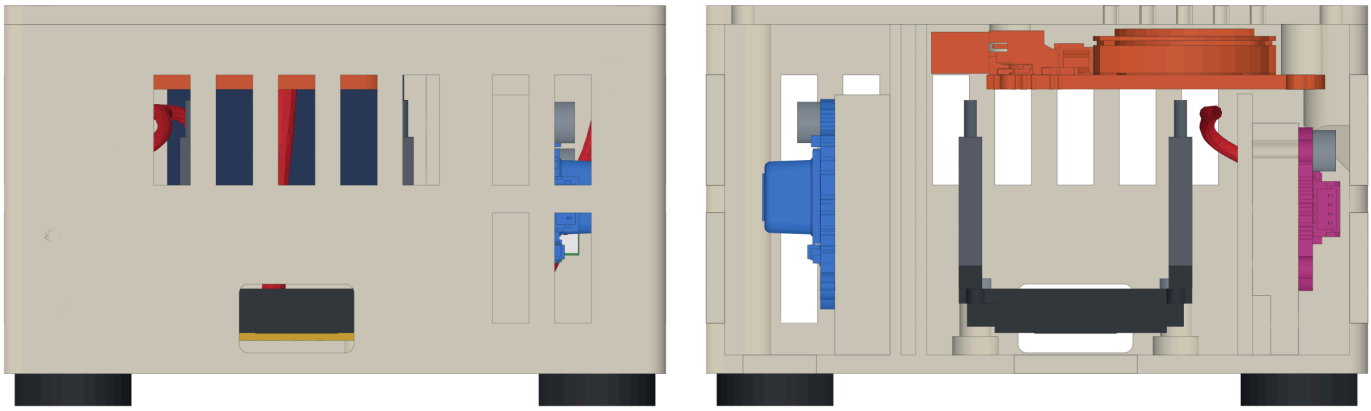


Figure 11 – Left: the back, seen from outside (column-j side on the left): one USB-C opening big enough for the plug's grip, the ESP32's 5 vent slots above it, and the chamber's 2 slot columns on the right. Right: cut across the case 30 mm from the back wall, seen from the front: the sensor chamber with the SCD-41, the double wall, the ESP32 upside down with its jumper plugs, the wall, and the SGP40 on the column-j wall; the speaker under the lid.

Printed parts

#	Part	Printed
1	**Base:** floor, walls, double wall with the cable hole, the wall to the column-j compartment, standoffs and posts for the sensor boards, ESP32 posts, light-sensor posts and window, OLED holder, LED sleeve, lid-screw bosses, vents, USB opening, display window, button hole	upright
2	**Lid** with the speaker grille, 4 speaker standoffs and 2 pegs that keep the display down	upside down

Both parts print without supports, apart from the short bridges over the front windows and the USB opening (see *Printing at school*). The feet are bought stick-on rubber feet, so the floor lies flat on the build plate.

Parts to buy for the case

Part	For	Notes
Waveshare ESP32-C6-DEV-KIT-N16, with headers (Berrybase WS-30547, €9.90)	the ESP32 in the case	Same pin rows as the DevKitC-1 (see *Hardware notes*). 55 × 25.4 mm with 4 corner holes ⌀2.
Adafruit **BMP580** (6411)	air pressure for the SCD-41's pressure compensation	STEMMA QT, I²C 0x46 or 0x47; Adafruit BMP5xx library
Adafruit **VEML7700 right angle** (5378)	ambient light; dims display and LED at night	STEMMA QT, I²C 0x10, sees light along the board
Adafruit **SGP40** (4829), *later*	air quality (VOC index 0–500)	STEMMA QT, I²C 0x59. Uses the humidity and temperature of the SHT45. Its place in the case is ready.
Adafruit **STEMMA Speaker** (3885)	alarm beeps	class-D amplifier, 3–5 V, 3-pin JST PH input
Panel button, Berrybase 10130 (red)	wake and sleep	M7 × 0.75, 27.5 mm long, nut and washer included, solder lugs
8 mm addressable LED, Berrybase PL9823-F8 (diffused), and a **1N4148** diode	CO_2 colour on the front	WS2812-style: one data wire, no resistors. Rated 4.5–6 V, so it runs from 5 V; the diode in its supply (about 4.2 V left) makes the ESP32's 3.3 V data count as high. Colour order red–green–blue.
STEMMA QT cables, 50 mm: 3 (4 with the SGP40)	SCD-41–SHT45, SHT45–light sensor, light sensor–BMP580 (BMP580–SGP40)	you have 2
STEMMA QT cable with female sockets, 150 mm (Adafruit 4397)	ESP32 pins to the first sensor	or your STEMMA QT to male-pin cable plus 4 female–female jumper wires
Female–female jumper wires, about 10, 10–20 cm	button, LED and display wires	cut in half and solder the cut end to the part; heat-shrink over each joint
JST PH 3-pin cable with female sockets (Adafruit 3894)	speaker to the ESP32 pins	–
4 stick-on rubber feet, ⌀10–13 × 3–4 mm	under the base	for example 3M Bumpon SJ5012 (⌀12.7 × 3.5 mm). At least 3 mm tall, so air reaches the floor vents. Their middles go 7.5 mm in from the outer edges.
4 × M3 × 10	lid	self-tapping into the bosses
6 × M2.5 × 6 hex socket (ISO 4762), 8 with the SGP40	SCD-41, SHT45, BMP580 (SGP40), 2 each	cut their own thread in the standoffs; turned from outside through the slots
2 × M2.5 × 5 hex socket	light sensor	from above into its posts (3.6 mm of plastic)
3 × M2 × 6 button head (ISO 7380, head ⌀3.5)	ESP32	at the corner next to the 3V3 pin a socket head (⌀3.8) would touch that pin's plug
4 × M2 × 6	speaker, under the lid	–
Drills ⌀1.6, 2.1, 2.5, 3.2, 7 and 8 mm, a pin vise	opening the printed pilot holes	see *Printing at school*; the school workshop may have them
Precision screwdriver, 2 mm hex	side-slot screws	the blade must be thin (about ⌀2.3 mm) to reach through the 4 mm slots; a Phillips PH1 is too thick

Dust mesh	behind the chamber's side slots	keeps dust out, lets gases through; type and thickness still open

Pins in the case

Part	ESP32 pins (row-pin)	Note
I^2C chain	3V3 (J1-1), GND (J3-1), SDA GPIO23 (J3-5), SCL GPIO22 (J3-6)	the 150 mm cable with female sockets
Panel button	GPIO2 (J1-12), GND (J3-15)	as now: input with pull-up, pressed = low
Speaker	SIG GPIO3 (J1-13), + 5V (J1-14), − GND (J1-15)	three pins next to each other; tones by PWM
8 mm LED	DIN GPIO18 (J3-10), VDD 5V through the diode, GND (J3-12)	the 5V pin feeds the speaker too: join the two 5 V wires

J1 and J3 are the two pin rows as in section 3; the Waveshare board has the same rows. Upside down in the case, J1 lies along the double wall and J3 along the column-j side. None of these is a strapping pin. Still to do in the firmware: the CO_2 colour moves from the on-board RGB LED (GPIO8, now facing the floor) to the LED on GPIO18, with red and green swapped for the PL9823; the speaker, BMP580, light sensor and SGP40 aren't in the firmware yet.

Cables and the I^2C chain

Without the breadboard there is no bus row, and the ESP32 has only one 3V3 pin, so all I^2C parts hang on **one chain** of STEMMA QT cables:

ESP32 → SGP40 → BMP580 → light sensor → (hole in the double wall) → SHT45 → SCD-41

- **ESP32 to the first sensor:** a 150 mm cable with female sockets on the four pins. It runs along the ESP32's side of the wall and round its back end. Until the SGP40 is fitted, it goes on to the BMP580's back socket; later the SGP40 goes in between with one more 50 mm cable, and nothing else changes.
- **All other links** are 50 mm cables. The model checks each one for length: the two short runs in the bay have 17–23 mm to spare, which forms a small loop; the one in the chamber loops up towards the outer wall.
- **The OLED** has only one pin header, so it can't sit in the middle of the chain. Its 4 wires are soldered to the light sensor's header pads (VIN, GND, SCL, SDA), which lie flat at the sensor's back edge, and end in female plugs on the OLED's pins.
- **Addresses:** OLED 0x3C, SCD-41 0x62, SHT45 0x44, BMP580 0x46/0x47, VEML7700 0x10, SGP40 0x59: no clashes.

Design rules behind it

- Sensirion's design-in rules: sensors close to large openings with little dead air around them, away from heat sources (the ESP32 module, its voltage regulator and the USB chip), the temperature sensor low because warm air rises.
- The SHT45 sits 8.5 mm from the SCD-41, never above it, at the front, farthest from the ESP32.
- The speaker is far from the SCD-41, which measures acoustically and doesn't like vibration.
- Walls, floor and lid 2 mm; double wall 2 × 1.2 mm with a 1.6 mm air gap; the wall to the column-j compartment 1.6 mm; all vents are 4 mm slots (air gets in, fingers don't).
- Every side-slot screw sits behind a slot, and every cable has its real length; the model checks both.

Sensor mounting and side slots

The sensor boards are **screwed on**. Each board stands on two **standoffs**: pillars that rise from the floor as part of the base, in the chamber against the double wall, in the column-j compartment against the new wall. Two **M2.5 × 6 hex socket screws** go through the board's holes on the edge away from its header; that edge is at the top.

A screwdriver doesn't fit inside the chamber. It is 18 mm wide, and only 8.6–11.6 mm is left in front of a screw head, while a 1/4" bit is 25 mm long and the short leg of a 2 mm hex key about 16 mm. So the screws are turned **from outside, through the side slots**, with a 2 mm hex precision screwdriver. The screw heads don't fit through a slot: with the lid off, push the screws through the board, hold the board on its standoffs from above and tighten them from outside. The column-j compartment works the same way, from the other side of the case.

Board	Size	Height overall	Mounting holes	Used
SCD-41	25.4 × 22.9 mm	7.9 mm (sensor)	4 × ⌀3.0 mm	the 2 at the top
SHT45	25.4 × 17.8 mm	4.5 mm (sockets)	4 × ⌀2.5 mm	the 2 at the top
BMP390 / BMP580	25.4 × 17.8 mm	4.5 mm (sockets)	2 × ⌀2.5 mm, on the edge opposite the 8-pin header	both
SGP40	25.4 × 17.8 mm	4.5 mm (sockets)	4 × ⌀2.5 mm	the 2 at the top

From Adafruit's board files and 3D models; the two agree on the outlines and holes. The holes are 2.54 mm from the edges, so the two used holes are 20.32 mm apart on every board. All parts sit on the front; the back is flat. Adafruit's product pages give slightly different heights (SCD-41 7.7, SHT45 4.8 mm): rounded caliper values, like the 25.5 mm they give for these 25.4 mm boards.

Parts on the boards, from Adafruit's 3D models (data sheet values in brackets):

- **Boards** are 1.57 mm thick (the usual “1.6 mm”).
- **SCD-41 sensor:** 10.1 × 10.1 mm, 6.3 mm above the board (0.8 mm base plus 5.5 mm cap, each ±0.3 mm; Sensirion's short summary says 6.5 mm). It sits in the middle of the board's width, its centre 12.2 mm from the header edge. The cap narrows to 7.8 mm at the top, where the white membrane is.
- **STEMMA QT sockets** (JST SH, 4 pins): 3.0 mm above the board, 6.0 mm wide, 4.95 mm deep, one in the middle of each short edge. Their openings sit 0.1–0.7 mm in from the board edge, so a plugged-in plug sticks out past the edge (S3 measures how far).
- **SHT45, BMP580 and SGP40 chips** are tiny (1.5 × 1.5 × 0.5, 2.0 × 2.0 × 0.8 and 2.4 × 2.4 × 0.9 mm), so on those boards the sockets stick up furthest.
- **Right-angle VEML7700 (5378):** the sensor (6.8 × 2.35 × 3.0 mm) lies 2.5 mm high on the board, its window 0.1 mm in from the edge opposite the header; the board is 4.5 mm tall overall.
- **STEMMA speaker (3885):** board 36.8 × 30.5 × 1.57 mm, 6.6 mm tall overall with the speaker (the product page says 7.2 mm).

3D models: Adafruit's STEP models of the SCD-40 (same board outline and sensor as the SCD-41), SHT45, BMP390, SGP40 (added 4 Nov 2026), right-angle VEML7700 and STEMMA speaker boards are in `cad/adafruit/`, from Adafruit's *Adafruit_CAD_Parts* collection on GitHub (MIT licence). They go into the case model to check the fit. The BMP390 model stands in for the BMP580: the outline and holes are the same, and both chips are about 2 × 2 mm.

Side slots: vertical slots 4 mm wide in two equal rows 12 mm tall (z 5.5–17.5 and 20.5–32.5), the columns 6.77 mm apart, a third of the 20.32 mm hole spacing, so both screws of a board sit behind a slot.

- **Chamber side:** 9 columns, centred on the wall. The SCD-41's screws sit behind columns 1 and 4, the SHT45's behind 6 and 9. The chamber's back has 2 columns; its front (the display side) has none.
- **Column-j side:** 9 columns on their own spacing, starting 9.5 mm from the back wall, so that all four screws of the SGP40 (columns 1 and 4) and the BMP580 (6 and 9) sit behind a slot. The two boards are 8.5 mm apart, room for the plugs and the loop of the cable between them.
- **ESP32:** its side of the bay has no side slots any more (the column-j slots now open into the compartment). It gets 6 floor vents under the board and 5 slots in the upper row of the back wall, round the USB port. The ESP32, its

voltage regulator and the USB chip turn about 0.4–0.6 W into heat; cool air comes in from below and the warm air leaves high at the back.

Dust mesh: the slots let in a little dust, so a mesh sits behind the chamber's slots in a groove on the inside of the outer wall. It slides in from above, so with the lid off it can be pulled out whenever the screwdriver is needed.

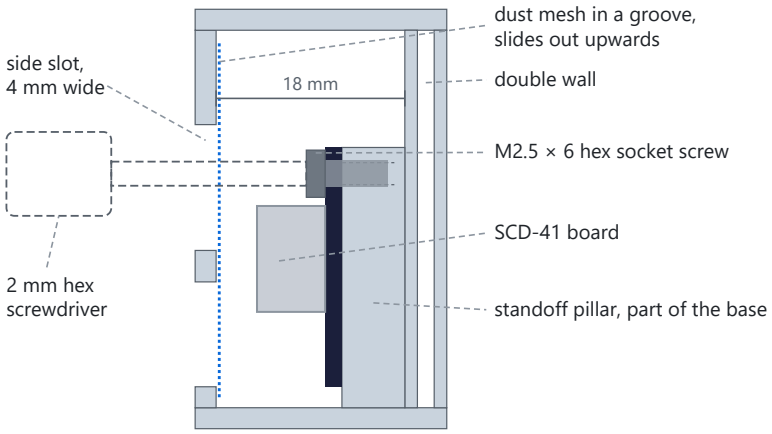


Figure 12 – Cut across the sensor chamber through a sensor screw, seen from the front, to scale. Dashed: the path of the screwdriver during assembly, with the lid off and the dust mesh pulled out. The column-j compartment is built the same way, mirrored.

Still open.

- **Waveshare board, underneath:** the module's position and the height of the parts under the board are read from Waveshare's photo. Check them when the board is here: the module should clear the 4th post by about 0.3 mm, and nothing under the board may be taller than the USB-C port (3.3 mm).
- **ESP32 screw at the 3V3 corner:** the button head leaves about 0.1 mm to the 3V3 pin's plug. If it touches, leave that screw out; the two at the USB end hold the board.
- **LED size:** there is no data sheet for the PL9823-F8, so the model uses Adafruit's 8 mm LED (11 mm tall, rim ⌀8.9). Measure it (P2) before printing: the sleeve's length sets how far the dome stands out.
- **Dust mesh:** type and thickness (P5), which set its groove. The groove isn't modelled yet.
- **OLED board thickness** (O3) isn't measured yet: the holder assumes 1.2 mm with 0.3 mm play.
- **Cable hole** in the double wall: seal it after wiring, for example with a bit of foam or putty.

Measurements for the 3D model

The printable model is built from these numbers. The Adafruit boards' outlines and holes come from their board files, but how the parts sit on the breadboard, the OLED and the cables have to be measured. The IDs match Figure 13.

v4. These measurements were taken for the v3 design. Without the breadboard and with the Waveshare board, B1–B4, E1–E5 and W1 are no longer used; the OLED (O), sensor-board (S) and part (P) values still are. New to measure when the parts are here: on the Waveshare board, the gap from the module to the 4th hole, the tallest part under the board and the middle of the USB-C port; the LED (P2); and the light sensor's height.

- **Tool:** a digital caliper, read to 0.1 mm. Measure each value twice; if the two differ by more than 0.2 mm, measure a third time. Don't squeeze soft parts such as the adhesive pad or the cables.
- **Leave the breadboard as it is.** Measure with the ESP32 and the wires plugged in, the way they will sit in the case. All heights are from the breadboard's top surface.
- **Photos** let me check the numbers. Take them straight on from 30–50 cm away and zoom in (that keeps the edges straight), with a ruler lying at the same height as the part. Wanted: the breadboard from above, from the column-a side and from the USB end; the OLED from the front (switched on) and from the back.

- **So far** is the value the drawings use now, from a shop page, a data sheet or a guess. A few tenths of a millimetre off is normal; a bigger difference changes the model.

ID	What to measure (mm)	So far	Measured
Breadboard (Berrybase, 400 contacts)			
B1	Length (row-1 end to row-30 end) and width (column-a edge to column-j edge), on the body without the latches	83 × 55	84.5 × 55.7 including the joining nubs
B2	Height from the table to the top surface, with the adhesive pad as it is now. Is the backing paper still on?	8.5 (guess)	9.3 without parts
B3	Side latches (the tongues and grooves for joining boards): on which edges, how many, how far they stick out, how wide. A photo from above is enough.	–	on the row-1 (USB) end and the column-a side; included in B1 (body 83 × 55 by the shop)
B4	From the row-1 end edge to the middle of the row-1 holes, and from the column-a edge to the middle of the column-a holes	–	
ESP32 on the breadboard			
E1	Gap between the breadboard top and the underside of the ESP32 board (the black plastic of its pin headers)	2.5 (usual header)	
E2	Highest point of the ESP32 board above the breadboard top (usually the metal cover of the module)	–	
E3	For each USB-C port: height of its middle above the breadboard top, and distance from its middle to the column-a edge of the breadboard	5.7; USB 20.3, UART 34.7	top of the USB-C ports 16.5 above the breadboard's *bottom*
E4	How far the USB ports stick out past the breadboard's row-1 edge (a minus value if they sit behind it)	0 (flush)	0 in rows 4–19: flush with the body, the nubs reach about 1.5 further. Now in rows 3–18: about 2.5 past the body, 1 past the nubs.
E5	Your USB-C cable: width × thickness of the plug's plastic grip, and its length up to where the cable can bend	–	not measured: different cables get used
Wires and cables			
W1	Highest point of any wire or cable above the breadboard top, after pressing them down as flat as they go. Where is it (row and column)?	–	30.0 above the breadboard's *bottom*, at rows 27–30: the STEMMA QT to pins cable
W2	STEMMA QT cables you have: how many plug-to-plug and plug-to-pins cables, and how long each one is	50 plug–plug	only 50 mm plug–plug; 50 mm is the shortest ready-made length (Adafruit, Pimoroni, SparkFun)
OLED (Joy-it SBC-OLED01)			

O1	Board width × height × thickness	–	27.75 × 28; about 12 deep with the pins
O2	Glass: width × height, and its distance to each board edge (left, right, top, bottom)	–	26.75 × 19.15, 1.8 thick; photo: 0.5 in from left and right, top edge 4.4 below the board's top edge
O3	Total thickness from the back of the board to the front of the glass. Any parts on the back, and how tall?	–	
O4	Lit area: where the pixels are, as distances from the glass edges. Easiest from the front photo with the display on.	21.7 × 10.9 (usual 0.96" panel)	photo: about 21.7 × 10.9, 2.5 in from the glass's left and right edges, 2.1 below its top edge
O5	Mounting holes: diameter, and the distance of their middles from the board edges	–	⌀1.5 (caliper); photo: middles 1.9 from the sides and 2.1 from top and bottom (23.9 apart)
O6	Pin header: straight or angled; how far the pins stick out behind the board, including the black plastic	–	straight; black plastic and pins at the back. Solder ends on the glass side stick out 1.8, level with the glass; pin row 1.5 below the top edge, centred
O7	Which glass edge has the thin flex strip that folds round to the back (the case must not press on it)	–	bottom edge, opposite the header: the lowest 4.2 of the glass is the connector strip; the flex folds round to the back through a notch about 14.5 wide and 2 deep
Sensor boards (SCD-41, SHT45)			
S1	Board thickness. Also the overall height of each board, from its back to its highest point.	1.57; overall SCD-41 7.9, SHT45 4.5 (3D model)	SCD-41 25.5 × 22.75 × 8; SHT45 25.5 × 17.75 × 4.75 overall
S2	Tallest part, from the board's front surface: on the SCD-41 the sensor's metal cap, on the SHT45 a STEMMA QT socket	SCD-41 6.3; SHT45 3.0 (3D model)	
S3	With a STEMMA QT cable plugged in: from the board edge to where the cable can bend by 90° without forcing it	–	
S4	Pin headers soldered onto any sensor board? If yes, which one, and how far they stick out at the back	none	
Parts for the case (only the ones you already have)			

P1	Panel button 10130 (v3: 10131, same size): usable thread length; cap diameter and height in front of the panel; depth behind the panel including the solder tags; nut across flats and thickness; washer outer diameter and thickness	product sheet: M7 × 0.75, thread 6.3 usable, knob 5.5 above it, collar ⌀9.7, 15.7 behind the collar; nut 10.1 across corners × 1.7; washer ⌀9.3 × 1.1	28 long, ⌀9; thread ⌀6.7; washer ⌀9
P2	LED (PL9823-F8): body diameter, rim diameter at the base, height from the rim to the top of the dome	8; rim 8.9; 11 (Adafruit's 8 mm LED)	not bought yet
P3	LED holder	–	not needed in v4: the LED sits in a printed sleeve
P4	2 mm hex screwdriver: blade diameter, blade length from the tip to the handle, handle diameter	blade 2.3	
P5	Dust mesh: material and thickness	–	not bought yet
P6	STEMMA speaker 3885: length × width × height	36.8 × 30.5 × 6.6 (3D model)	37 × 32.8 × 7.3; the 32.8 includes the cable at the side

First measurements entered on 1 Nov 2026; the empty cells are still open. Values marked *photo* come from Joy-it's front photo of the OLED, scaled to the measured board size (±0.3 mm); the lit area matches the usual 0.96" panel. The pressure sensor and the light sensor aren't bought yet. Their sizes come from Adafruit's 3D models and board files; check the board size, holes and tallest part when they arrive.

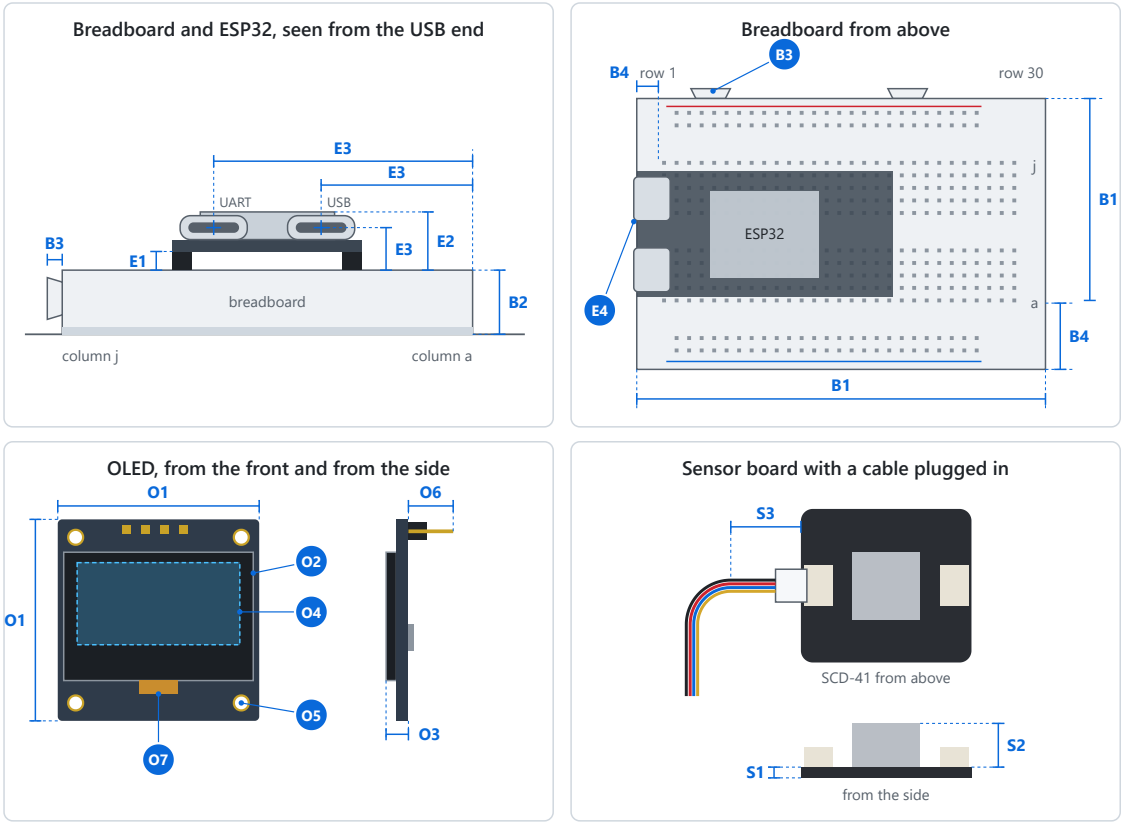


Figure 13 – Where to measure (sketches, not to scale). Top left: breadboard and ESP32 seen from the USB end, column a on the right. Top right: the breadboard from above, row 1 (USB end) on the left as in Figure 9. Bottom left: the OLED from the front and, beside it, from the side (glass on the left, pins on the right). Bottom right: a sensor board from above with a cable plugged in, and from the side. Blue: the IDs from the table.

Also still needed.

- **Screws:** M3 available for the lid? Otherwise the lid gets snap hooks.
- **Dust mesh:** type and thickness (P5), which set the size of its groove.

Printing at school

The case is printed on the school's 3D printer. The teacher's answers (2 Nov 2026):

Question	Answer
Printer	**Stratasys F370**, an industrial FDM printer. Its build space is 355 × 254 × 355 mm, so both parts fit easily.
Material	**ABS or ASA**, in **black or ivory**, whatever is in stock. No clear or translucent material.
Slicer	**GrabCAD Print**, Stratasys' own slicer; the printer type is set there.
Layer height	**0.254 mm**, the standard setting.
Holes	Print them **smaller, as pilot holes**, and **drill them to size** afterwards. Then the size is right and the holes are round.
Files	**STEP** or native CAD files, or a ready file from GrabCAD Print. The teacher prints the part and reports back. **Free of charge.** Test pieces are welcome.

What this means for the model:

- **Files:** each part (base, lid) as its own STEP file, in millimetres. STEP keeps exact circles and the units; an STL is only a mesh of triangles. So the CAD program must be able to write STEP. A ready GrabCAD Print file would need that

program and a GrabCAD account, and the school sets the printer anyway, so STEP is the simpler way. With the files goes a note: base upright, lid upside down (as in *Printed parts*), and the colour wish.

- **Accuracy:** Stratasys gives ±0.2 mm for the F370 (±0.2 % on parts longer than 100 mm), so about ±0.2 mm on this case. The fit gaps (OLED holder, USB opening, slots) are set in one place in the model and checked with the test piece.
- **Layers:** 0.254 mm. Floor and lid thickness and steps in height are best whole layers, for example 2.03 mm (8 layers) instead of 2.0 mm. Details thinner than about 2 layers, such as a 0.5 mm skin over the light sensor, don't print reliably.
- **Supports:** for ABS and ASA the F370 prints soluble support, which is washed out in a heated bath afterwards. GrabCAD Print adds it wherever the model overhangs, also in narrow gaps like the double wall's 1.6 mm air gap, where it is hard to wash out. So the parts stay designed without overhangs. The exceptions are the flat tops of the windows in the front wall (OLED, light sensor, USB). They are short bridges through open holes, so any support there washes out easily.
- **Colour:** in direct sun a black case warms up more than an ivory one, and the SHT45 would read that warmth; out of the sun it makes no difference. Neither colour lets light through, so the light sensor gets an open window.
- **Test piece first:** a small piece with the critical fits: a standoff behind a 4 mm slot with its pilot hole (drill it, then screw a board on), a lid-screw boss, the button hole, the LED sleeve, an ESP32 post, the USB-C opening and the OLED window. Then the full case.

Hole	Printed as	Drill to	For
Sensor standoffs in the base, 8, horizontal (chamber and column-j compartment)	⌀1.6, teardrop with the point up as printed (no support inside)	⌀2.1	M2.5 × 6 screws cut their own thread. Drill them from outside through the side slot, before the boards go in.
Light-sensor posts, 2, vertical, through the floor	⌀1.6	⌀2.1	M2.5 × 5 screws from above
ESP32 posts, 3, vertical	⌀1.2	⌀1.6	M2 × 6 button-head screws from above
Speaker standoffs on the lid, 4, vertical	⌀1.2	⌀1.6	M2 × 6 screws, from below through the board
Lid-screw bosses in the base, 4	⌀1.8	⌀2.5	M3 lid screws, self-tapping
Lid, 4	⌀2.5	⌀3.2	the M3 screws pass through
Panel button, front wall	⌀6.0	⌀7.0	M7 × 0.75 thread (measured ⌀6.7, P1)
LED, front wall and sleeve	⌀7.5, teardrop with the point up	⌀8.0	drill from the front through wall and sleeve in one go; the LED is pushed in from behind until its rim stops

Pilot holes are 0.4–1 mm smaller than the final size, so the drill cuts all round and the hole ends up round. Drill the small ones by hand with a pin vise and the 7 and 8 mm ones slowly: ABS and ASA melt and smear when the drill runs hot. The sizes are checked with the test piece.

3D model

The printable model is a Python script, `cad/case_v4.py`, written with **build123d** 0.13.0. build123d is a free CAD library (Apache 2.0) built on OpenCascade, the same geometry kernel FreeCAD uses. Every size is a named number at the top of the script, in these coordinates:

- **x** runs from the inside of the back wall (the USB end) towards the front.
- **y** runs from the bay side of the double wall; plus is the electronics bay towards the column-j wall, minus the sensor chamber.
- **z** runs up from the underside of the floor.

To change the case, change a number and run the script again. It writes STEP files, as the school wants, and checks the fit itself. The v3 model (breadboard in the case) stays in `cad/case.py` and still builds into `cad/out/`.

- **Setup, once** (PowerShell in the project folder): `& "C:\Program Files\Python313\python.exe" -m venv cad\.venv`, then `cad\.venv\Scripts\python.exe -m pip install -r cad\requirements.txt`. This puts about 60 packages into `cad\.venv`. build123d runs on Python 3.11–3.14. Use the installed 3.13, because the `py` launcher's 3.14 entry is broken on this PC.
- **Build:** `cad\.venv\Scripts\python.exe cad\case_v4.py` (about 60 s). It writes `cad/out/v4/`: `base.step` and `lid.step` for the school, an STL of every part for Cura, and `viewer.html`.
- **Viewer:** double-click `cad/out/v4/viewer.html`. It shows the case with Adafruit's models of the sensor boards and the speaker, the plugs and cables, the OLED drawn from the measurements (O1–O7) with its jumper-wire plugs, and simple models of the ESP32 with its plugs, the button with its nut, the LED, the screw heads and the rubber feet. Parts fixed to the lid lift with it. You can turn the model, switch parts on and off, lift the lid or make it see-through, and cut the case open along x, y or z. A view can also be set in the address, for example `viewer.html#iso&lift=30&hide=lid,cables` (also `isoBack`, `zoom`, `at=x,y,z`, `cut=y,-12`, `ghost`, and `clean` for pictures without the panel; Figures 8–11 were made that way). It needs an internet connection, because it loads the three.js library.
- **Checks:** the script prints each part's size and volume and whether it is a valid solid. It also runs four fit checks:
  - No board, plug, cable or bought part overlaps the case.
  - No two of those parts overlap each other.
  - The 2.3 mm screwdriver blade reaches every side-slot screw through a slot.
  - Each 50 mm cable is checked against its real length, assumed to be 41 mm of wire between the plugs; the loop between two facing plugs is grown until it has that length.

  Now: the base is 86.3 × 72 × 38 mm, and the lid 2.03 mm thick plus the speaker standoffs. Nothing overlaps apart from the button's thread and the LED in their printed pilot holes, which are drilled out. All 8 side-slot screws are reachable. The chamber cable's loop stays 2.6 mm under the lid.

In the model:

- **Base:**
  - Walls, floor and the double wall with its air gap. At the bottom of its front end, a hole with a pointed top lets the cable from the light sensor and its plug into the chamber; seal it after wiring.
  - The wall to the column-j compartment, 1.6 mm thick, against the BMP580's and SGP40's posts.
  - Standoffs for the SCD-41 and SHT45, posts for the BMP580 and SGP40, all with ⌀1.6 teardrop pilot holes.
  - 4 ESP32 posts: a ⌀5 foot and a ⌀3.4 neck that fits between the parts under the board; 3 with ⌀1.2 pilot holes, the 4th cut flat on the module's side.
  - 4 light-sensor posts, 2 with ⌀1.6 pilot holes through the floor, and its 7.5 × 3.1 window in the front wall.
  - The OLED holder: the recess for its glass, two rails and two shelves on the front wall.
  - The LED sleeve on the floor behind the front wall, with its ⌀7.5 pilot hole.
  - The 4 lid-screw bosses, each filling an inside corner, with ⌀1.8 pilot holes. The two on the column-j side hang from the walls with 45° undersides: the back one above the SGP40's plug, the front one above the button.
  - The 4 mm slots, the floor vents and the back-wall slots; the USB opening, the OLED window and the button's ⌀6.0 pilot hole.
- **Lid:** the speaker grille, 7 × 5 holes of ⌀1.8 over the speaker. ⌀2.5 pilot holes for the screws. Four speaker standoffs with ⌀1.2 pilot holes. Two pegs that keep the OLED down. No vents.
- **Bought parts** (not printed):
  - The SCD-41 is drawn with Adafruit's SCD-40 model (same board), and the BMP580 with the BMP390 model (same board and holes). The SGP40, the VEML7700 5378 and the speaker 3885 are Adafruit's models too.

- The Waveshare ESP32 is drawn from Waveshare's drawing: board, holes, pin rows, the module, USB-C port and buttons underneath (from the photo), and the jumper plugs on the pins it uses.
- The OLED is drawn from the measurements; the button from its product sheet; the LED from Adafruit's 8 mm LED.
- The STEMMA QT plugs and cables, the display's wires and the speaker's JST PH plug are drawn too.

Not in the model yet: the dust-mesh groove, and the button, LED and speaker wires. `cad/out/` and `cad/.venv/` are not stored in git; the setup and the scripts make them again.

14 Change log

Version	Date	Changes
0.20	4 Nov 2026	Case (section 13) is now **design v4**, a compact case without the breadboard: 86.3 × 72 × 40 mm instead of 105 × 86 × 40 mm, same look. New model `cad/case_v4.py` (writes into `cad/out/v4/`); `cad/case.py` keeps v3 for reference. • The Waveshare ESP32-C6-DEV-KIT-N16 (to be bought) lies upside down on 4 posts at the back, its USB-C low in the back wall; jumper-wire plugs sit on its pins. New hardware notes for it (section 11). • The sensor chamber holds only the SCD-41 and SHT45. A wall separates a column-j compartment from the ESP32, with the BMP580 and room for an SGP40 air-quality sensor later, both screwed on through their own side slots. The ESP32 gets floor vents and slots high in the back wall. • The light sensor lies on 4 posts with 2 screws; the display sits 1.6 mm higher. Panel button Berrybase 10130 with its nut; an 8 mm addressable LED (PL9823-F8) in a printed sleeve below it. The speaker hangs under the lid over the ESP32. • One I²C chain of STEMMA QT cables, ESP32 → SGP40 → BMP580 → light sensor → SHT45 → SCD-41; the OLED is wired to the light sensor's pads. New tables *Pins in the case* and *Cables and the I²C chain*; new parts list. • Figures 8–11 are now pictures of the 3D model; the v3 drawings are gone. Updated: side slots, still-open points, measurement notes (P1–P3), pilot-hole table, *3D model*. Adafruit's SGP40 model added to `cad/adafruit/`. The viewer page has new address options `isoBack` and `clean`. Considered and not taken: Adafruit's 1.5" 128×128 OLED (case about 59 mm tall) and the 1.3" 128×64 OLED FeatherWing (about 94 mm wide); the 0.96" Joy-it display stays.
0.19	2 Nov 2026	Case (section 13), 3D model: the light-sensor window has the shape of the sensor's face (7.5 × 3.1 mm) instead of a round hole. The front face has no vents any more. The column-j side keeps only the 8 slot columns beside the ESP32 (proposal; new paragraph *Fewer vents* on why some are needed). The display holder is reworked without changing the outside: the glass sits in a 1 mm recess, open at the top, so the screen is 1 mm behind the outside instead of 2 mm, and two pegs under the lid keep the display from lifting. *Printing at school*: the flat-topped windows may get support, which washes out of the open holes. Figures 8–10 updated to match (front vents gone, column-j slots, light-sensor window).
0.18	2 Nov 2026	Case (section 13), 3D model: • The BMP580 is back in the base: on standoffs at the back of the chamber, turned on its side, so both its screws sit behind one slot column. The lid lifts off without unplugging anything. • The right-angle VEML7700 (5378) lies under the OLED again, in a pocket in the floor with its back edge under the breadboard, and looks out through a ⌀3.5 mm hole. The 4162 under the lid and its lid window are gone. • The cable opening is now a hole at the bottom of the double wall, wide enough for a STEMMA QT plug, instead of a notch from the top. • The lid vents are gone; the electronics vent through the column-j side wall. The panel button moved up to the middle of the display, and the RGB LED goes below it. • New in the model: the OLED with a holder on the front wall (two rails, two shelves), and the speaker on 4 lid standoffs under a grille that now sits over the speaker. Layout, light sensor, display holder and speaker holder are proposals (*Still open*). Figures 8–10: lid vents removed, button, LED, light sensor, cable hole, sensors and screw bosses moved. Parts list: 6 M2.5 screws (was 8) and 4 new M2 screws for the speaker, a ⌀1.6 drill. The viewer page takes a view from its address.

0.17	2 Nov 2026	Case (section 13): the 3D model now holds the sensors with their plugs and cables. This is a proposal, to be confirmed (*Still open*): • SHT45 at the front and SCD-41 in the middle on base standoffs, every screw behind a slot. BMP580 hanging from the lid at the back. • VEML7700 4162 under the lid, looking up through a tapered hole. The 5378 is back in stock but no longer fits under the OLED. • A cable notch in the double wall at the front. Both 50 mm cables are checked for length. Lid screws are now in the 4 corners, each boss filling its corner; the one at the back of the column-j side hangs above the breadboard. The vents are uniform: the same 4 mm slots in two equal rows on both side walls (13 columns, centred), 2 columns at the front and back of the chamber, and 11 floor vents in line with them. Parts list: 8 M2.5 screws, two 50 mm cables and a second STEMMA QT to male-pin cable.
0.16	2 Nov 2026	Case (section 13): new subsection *3D model*. The case is now modelled with build123d (Python). `cad/case.py` writes `base.step` and `lid.step` for the school, STL files and a 3D viewer page into `cad/out/`. The model has the shell, double wall, breadboard pocket, screw bosses, feet, vents, USB opening, OLED window and button hole. The parts that wait for open points are listed. Measured values moved the breadboard 1.5 mm forward and its top 0.8 mm up. The USB opening is now one piece, 0.7 mm higher, with 1 mm corners. The printed feet are replaced by 4 stick-on rubber feet (new row in the parts list). The base now lies flat on the build plate and needs no support under the floor, with 13 floor vents between the feet. New files in the project files table (section 1).
0.15	2 Nov 2026	Case (section 13): new subsection *Printing at school* with the teacher's answers: Stratasys F370, ABS or ASA in black or ivory, GrabCAD Print, 0.254 mm layers, STEP files, free of charge, test pieces welcome. It lists what follows for the model (STEP export, ±0.2 mm accuracy, whole layers, no overhangs because of the soluble support, no clear material, a test piece first) and a table of pilot holes and the drill sizes they are opened up to. Drills and a pin vise added to the parts list. The printer question is gone from *Also still needed*.
0.14	1 Nov 2026	Case (section 13): the list at the end is now a measurement sheet, *Measurements for the 3D model*. It has 27 values with IDs (breadboard, ESP32, wires, OLED, sensor boards, case parts), how to measure them, the value the drawings use so far and an empty column to write in. New Figure 13 shows where each value is measured. The printer, screw and mesh questions are in a box below it. Adafruit's 3D models of the SCD-40, SHT45, BMP390, right-angle VEML7700 and speaker boards are now in `cad/adafruit/` (MIT licence). The sensor-board table gives each board's overall height from them, and a new list gives the parts on the boards: boards 1.57 mm thick, SCD-41 sensor 10.1 × 10.1 × 6.3 mm, STEMMA QT sockets 6.0 × 4.95 × 3.0 mm. First measurements entered (breadboard, USB height, wires, OLED, SCD-41, SHT45, panel button, speaker), with a new row P6 for the speaker. OLED glass, lit area, holes and flex strip read from Joy-it's photo. Breadboard height, nub positions, USB position, the tallest cable and the OLED hole size added. Wiring (section 4): the ESP32 moved one row towards the USB end, to rows 3–18, so its USB-C ports reach past the breadboard's joining nubs; all its holes moved by one row (for example 3V3 in i18, GND a18, SCL a13, SDA a14, GPIO2 j7), and so did the planned RGB LED (a9–a11) and speaker (a12) holes. The GPIO numbers and the firmware are unchanged.
0.13	31 Oct 2026	Case (section 13): the sensor boards are screwed on with M2.5 × 6 hex socket screws to standoffs on the double wall; the 3 sensor clips are gone. The chamber's side vents are now 4 mm vertical slots in two rows, so a 2 mm hex screwdriver reaches the screws from outside, with a removable dust mesh behind them. New subsection *Sensor mounting and side slots* with the boards' mounting holes and Figure 12. BMP580 (6411) added as the in-stock pressure sensor. New *Still open* notes: sensor layout and the 50 mm cables, mesh size, light sensor. The case is printed on the school's printer.

0.12	31 Oct 2026	Sleep mode (section 6): the RGB LED is off and the display shows a small moon, while measuring, history and the web page carry on. Switched by holding the button for 2 s (new settings `LONG_PRESS_MS`, `SLEEP_MESSAGE_MS`), by a switch on the web page (new address `/api/sleep`) or with `sleep on|off` in the Serial Monitor; remembered through restarts. New in section 11: the power LEDs that no software can switch off, and Figure 7, where tape may cover the SCD-41 and SHT45 boards (drawn from Adafruit's board files). The case drawings are now Figures 8–11. New test step 10 and troubleshooting rows. `OLED_BRIGHTNESS` in the settings table now shows the sketch's value, 10. Tested on the board: web switch, `/api/sleep`, Serial commands, restart.
0.11	29 Oct 2026	Web page: the graphs are now right below the live values, with a heading that names the time shown. New buttons *Last 1 / 3 / 6 / 12 / 24 h* above them; the range moves along with the time (they replace *Last 24 h*). On a phone the day box now comes before the calendar. New screenshot (Figure 6), updated 8.3, 8.5 and test step 9. Tested on the board, on a PC and at phone width.
0.10	29 Oct 2026	Step 3: web page served by the monitor (new section 8, files `web.h`, `web.cpp`, `web/`). Live values, a calendar with marks on days that have values, a time range of up to 24 h, graphs of CO_2, temperature and humidity that can be switched on and off, a CSV download between two dates, and Auto/Light/Dark themes. Fixed address via the router (8.2). Uses 60 KB more program flash; the history storage is unchanged. New test step 9 and troubleshooting rows. Sections 8–13 are now 9–14, and the case drawings are now Figures 7–10. Tested on the board and in the browser, on a PC and at phone width.
0.9	29 Oct 2026	30 s warm-up after start-up (new setting `WARMUP_TIME_MS`): the sensors measure, but no value is shown, printed on Serial or saved in the history, and the readings from this time are thrown away. The display counts down meanwhile, then stays on for 30 s with the first values. `status` shows the seconds left. Updated test steps 3 and 4 and the troubleshooting row. Tested on the board: the warm-up ended exactly 30 s after the SCD-41 started, the first CO_2 value followed 3 s later.
0.8	28 Oct 2026	Step 2: Wi-Fi (name and password in `secrets.h`, which isn't in git) and internet time. New section 7: the values are stored in the flash with LittleFS as 1-minute values (7 days), 5-minute values (90 days) and hourly values with min/max (about 3½ years), saved every 10 minutes. New Serial Monitor commands `status`, `files`, `csv` and `save`. Warning about the partition scheme and "Erase All Flash" in 5.4, new setup step 5.8, test step 8 and troubleshooting rows. Sections 7–12 are now 8–13, and the case drawings are now Figures 6–9. Tested on the board: saving, restart and firmware upload keep the history.
0.7	25 Oct 2026	Section 11 now shows the approved case design v3: side by side, sensor chamber along the column-a side behind a double wall, display, light sensor, RGB LED and panel button on the front, speaker under the lid. New drawings, parts list, planned pins (RGB LED GPIO18–20, speaker GPIO21) and the measurements needed for the printable model.
0.6	25 Oct 2026	New section 11: concept for a 3D-printed case (flat box, display deck over the OLED, vented sensor chamber behind an inner wall), with four drawings, a parts list and the measurements needed for the printable files.
0.5	25 Oct 2026	New setting `OLED_BRIGHTNESS` (display brightness 0–255, default 207). RGB LED default brightness lowered to 5. More about OLED burn-in in section 9.
0.4.1	25 Oct 2026	Added sensor handling (what can harm the sensors, magnets, touching), placement advice, and how to check and restore a contaminated SHT45 (salt test, heater decontamination) to section 9. Expanded the SCD-41 handling note (membrane, mechanical force, ESD).

0.4	25 Oct 2026	The on-board RGB LED (GPIO8) now shows the CO_2 level, dimmed (`LED_BRIGHTNESS` = 10). Its colour changes smoothly from green (below 800 ppm) through yellow (1000) to red (2000 and above). It also turns off a colour left over from the BlinkRGB test at start-up.
0.3.1	25 Oct 2026	The "UART" USB port's CP2102N chip needed the Silicon Labs CP210x driver on this PC (Device Manager Code 28). Added the install steps (5.5) and a troubleshooting row. Clarified that the J5 jumper must stay fitted.
0.3	25 Oct 2026	Bus rows reordered to match the OLED's pins (row 27 GND, 28 3.3 V, 29 SCL, 30 SDA). OLED now lies flat in a27–a30, readable with the pins at the top; `OLED_ROTATION` back to 0. ESP32 wired straight to the bus rows (5 wires, power rails no longer used); button moved to rows 25–27 with its GND leg in the GND row.
0.2	25 Oct 2026	New breadboard layout: ESP32 in rows 4–19 with the USB ports flush with the board edge, bus moved to rows 27–30, button to rows 24–26. New `OLED_ROTATION` setting. Added SCD-41 vs SCD-30 comparison and notes on the SCD-41's internal T/RH.
0.1	25 Oct 2026	Initial setup: ESP32-C6 + SHT45 + SCD-41 + OLED on breadboard; button wakes display for 30 s; I^2C scanner; optional FRC calibration tool; this documentation.